

TRIPLE QUANTUM SCHUBERT CALCULUS

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ABSTRACT.

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1. INTRODUCTION

2. DEFINITIONS AND STATEMENT OF RESULTS

2.1. Quantum double Schubert polynomials and triple quantum Schubert structure constants.

We fix $n \geq 1$ and write

$$x = (x_1, \dots, x_n), \quad y = (y_1, \dots, y_n), \quad q = (q_1, \dots, q_{n-1}).$$

For $1 \leq a < b \leq n$, set

$$q_{ab} := q_a q_{a+1} \cdots q_{b-1},$$

and use the cohomological grading $\deg q_i = 2$ (equivalently $\deg q_{ab} = 2(b - a)$).

Let s_i act on the y -variables by swapping y_i, y_{i+1} , and define the divided difference operators

$$\partial_i^{(y)} f := \frac{f - s_i f}{y_i - y_{i+1}} \quad (1 \leq i \leq n-1).$$

For $w \in S_n$, if $w = s_{i_1} \cdots s_{i_r}$ is any reduced word, set

$$\partial_w^{(y)} := \partial_{i_1}^{(y)} \cdots \partial_{i_r}^{(y)}.$$

Following Kirillov–Maeno [5], define the quantum elementary polynomials $e_j^{(q)}(x_1, \dots, x_k)$ by

$$\Delta_k(t \mid x_1, \dots, x_k) := \sum_{j=0}^k t^{k-j} e_j^{(q)}(x_1, \dots, x_k),$$

where

$$\Delta_k(t \mid x_1, \dots, x_k) = \det \begin{pmatrix} x_1 + t & q_1 & 0 & \cdots & 0 \\ -1 & x_2 + t & q_2 & \ddots & \vdots \\ 0 & -1 & x_3 + t & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & q_{k-1} \\ 0 & \cdots & 0 & -1 & x_k + t \end{pmatrix}.$$

Let $w_0 \in S_n$ be the longest permutation. The top quantum double Schubert polynomial is

$$\tilde{\mathfrak{S}}_{w_0}(x; y) := \prod_{i=1}^{n-1} \Delta_i(-y_{n-i} \mid x_1, \dots, x_i),$$

and for $w \in S_n$ we define

$$\tilde{\mathfrak{S}}_w(x; y) := (-1)^{\ell(w_0) - \ell(w)} \partial_{w w_0}^{(y)} \tilde{\mathfrak{S}}_{w_0}(x; y).$$

Its specialization

$$\tilde{\mathfrak{S}}_w(x) := \tilde{\mathfrak{S}}_w(x; 0)$$

is the quantum Schubert polynomial. In particular, at $q = 0$ these recover the ordinary double and single Schubert polynomials.

Given their stability property, we may let the number of variables be infinite, indexing the polynomials by S_∞ . As a module over $\mathbb{Z}[q][y]$, the quantum double Schubert polynomials form a basis of $\mathbb{Z}[q][x, y]$, and we may write

$$\tilde{\mathfrak{S}}_u(x; y) \tilde{\mathfrak{S}}_v(x; z) = \sum_{w, d} q^d c_{u, v}^{w, d}(y; z) \tilde{\mathfrak{S}}_w(x; y)$$

where $d = (d_1, \dots, d_{n-1})$ is a weak composition and $q^d = q_1^{d_1} \cdots q_{n-1}^{d_{n-1}}$. The coefficients $c_{u, v}^{w, d}(y; z)$ are the *triple quantum Schubert structure constants*, which are polynomials in y and z with integer coefficients. This product respects the grading where for each i , $\deg y_i = \deg z_i = 1$ and $\deg q_i = 2$. In particular, if $c_{u, v}^{w, d}(y; z) \neq 0$, then $\deg c_{u, v}^{w, d}(y; z) = \ell(u) + \ell(v) - \ell(w) + 2 \sum_i d_i$. Unlike the usual triple Schubert structure constants, they do not exhibit triangularity in Bruhat order, and indeed we may have $\ell(w) < \ell(u) + \ell(v)$. There is an ordering in which the quantum double Schubert polynomials are triangular, but we will not need this in full generality.

Remark 2.1 ($\overset{k}{\rightsquigarrow}$ and the quantum k -Bruhat order). qrtre

For $d = (0, \dots, 0)$, this is the ordinary triple Schubert calculus structure constant, and it was proved in [3] that these have the positivity property that they are polynomials in the differences $y_i - z_j$ with nonnegative integer coefficients. We will refer to this as *triple Schubert positivity*. We have the following conjecture:

Conjecture 2.2. *For all u, v, w and d , the structure constants $c_{u, v}^{w, d}(y; z)$ exhibit triple Schubert positivity.*

Example 2.3. We compute the product $\tilde{\mathfrak{S}}_{15423}(x; y) \tilde{\mathfrak{S}}_{1432}(x; z)$:

$$\begin{aligned} & q_2 q_3 (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{124635}(x; y) + q_2 q_3 (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{13452}(x; y) \\ & + q_2 q_3 (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{14253}(x; y) \\ & + q_2 q_3 \tilde{\mathfrak{S}}_{125634}(x; y) + q_2 q_3 \tilde{\mathfrak{S}}_{126435}(x; y) + q_2 q_3 \tilde{\mathfrak{S}}_{13542}(x; y) + q_2 q_3 \tilde{\mathfrak{S}}_{214635}(x; y) \\ & + q_2 q_3 \tilde{\mathfrak{S}}_{23451}(x; y) + q_2 q_3 \tilde{\mathfrak{S}}_{24153}(x; y) + q_2 q_3 \tilde{\mathfrak{S}}_{31452}(x; y) + q_2 q_3 \tilde{\mathfrak{S}}_{41253}(x; y) \\ & + q_2 (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{146235}(x; y) + q_2 (y_4 + y_5 - z_1 - z_2) \tilde{\mathfrak{S}}_{24513}(x; y) \\ & + q_2 \tilde{\mathfrak{S}}_{246135}(x; y) + q_3 \tilde{\mathfrak{S}}_{1724356}(x; y) + q_3 \tilde{\mathfrak{S}}_{261435}(x; y) + q_3 \tilde{\mathfrak{S}}_{35142}(x; y) \\ & + (q_2 q_3 + q_2 ((y_1 - z_2)(y_4 + y_5 - z_1 - z_2) + (y_4 - z_1)(y_5 - z_1))) \tilde{\mathfrak{S}}_{14523}(x; y) \\ & + ((y_4 - z_1)(y_5 - z_3) + (y_4 - z_1)(y_1 + y_2 - z_1 - z_2) \\ & + (y_5 - z_2)(y_5 - z_3) + (y_5 - z_2)(y_1 + y_2 - z_1 - z_2)) \tilde{\mathfrak{S}}_{25413}(x; y) \\ & + ((y_1 - z_2)(y_1 - z_3)(y_4 - z_2) + (y_1 - z_2)(y_1 - z_3)(y_5 - z_1) + (y_1 - z_2)(y_5 - z_1)(y_5 - z_2) \\ & + (y_1 - z_3)(y_4 - z_1)(y_5 - z_1) + (y_4 - z_1)(y_5 - z_1)(y_5 - z_2)) \tilde{\mathfrak{S}}_{15423}(x; y) \\ & + (q_2 + (y_1 - z_2)(y_1 - z_3) + (y_1 - z_3)(y_5 - z_1) + (y_5 - z_1)(y_5 - z_2)) \tilde{\mathfrak{S}}_{156234}(x; y) \\ & + (q_3 + (y_1 - z_1)(y_4 - z_1) + (y_5 - z_2)(y_1 + y_4 - z_1 - z_2)) \tilde{\mathfrak{S}}_{45123}(x; y) \\ & + (q_2 q_3 + q_3 ((y_1 - z_1)(y_1 + y_5 - z_2 - z_3) + (y_5 - z_2)(y_5 - z_3))) \tilde{\mathfrak{S}}_{15243}(x; y) \\ & + q_3 (y_1 + y_5 + y_6 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{162435}(x; y) + q_3 (y_1 + y_2 + y_5 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{25143}(x; y) \\ & + q_2 q_3 (y_1 + y_2 + y_5 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{12543}(x; y) + q_2 q_3 (y_1 + y_2 + y_5 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{21453}(x; y) \\ & + q_2 q_3 ((y_1 - z_2)(y_1 - z_3) + (y_1 - z_2)(y_2 - z_2) + (y_1 - z_2)(y_4 - z_1) \\ & + (y_1 - z_2)(y_5 - z_1) + (y_2 - z_1)(y_4 - z_1) + (y_4 - z_1)(y_5 - z_3)) \tilde{\mathfrak{S}}_{12453}(x; y) \\ & + (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{1742356}(x; y) + (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{461235}(x; y) \\ & + (y_4 + y_5 - z_1 - z_2) \tilde{\mathfrak{S}}_{35412}(x; y) + (y_4 + y_5 - z_1 - z_2) \tilde{\mathfrak{S}}_{45213}(x; y) \\ & + ((y_1 - z_2)(y_1 - z_3) + (y_1 - z_2)(y_5 + y_6 - z_1 - z_2)) \end{aligned}$$

$$\begin{aligned}
& + (y_1 - z_3)(y_4 - z_1) + (y_4 - z_1)(y_5 + y_6 - z_1 - z_2)) \tilde{\mathfrak{S}}_{164235}(x; y) \\
& + (y_1 + y_2 + y_5 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{256134}(x; y) + (y_1 + y_5 + y_6 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{165234}(x; y) \\
& + (y_1 + y_2 + y_4 + y_5 + y_6 - 2z_1 - 2z_2 - z_3) \tilde{\mathfrak{S}}_{264135}(x; y) \\
& + \tilde{\mathfrak{S}}_{1752346}(x; y) + \tilde{\mathfrak{S}}_{265134}(x; y) + \tilde{\mathfrak{S}}_{2741356}(x; y) \\
& + \tilde{\mathfrak{S}}_{356124}(x; y) + \tilde{\mathfrak{S}}_{364125}(x; y) + \tilde{\mathfrak{S}}_{462135}(x; y) \\
& + \tilde{\mathfrak{S}}_{561234}(x; y)
\end{aligned}$$

By inspection, we can see that the coefficients exhibit triple Schubert positivity.

We will provide formulas below that demonstrate this in special cases. In particular, we prove

Theorem 2.4. *Suppose $u, v, w \in S_\infty$ and v is k -Grassmannian for some k . Then for any d , $c_{u,v}^{w;d}(y; z)$ exhibits triple Schubert positivity.*

2.2. The parabolic case. Unlike ordinary cohomology or equivariant cohomology, quantum cohomology and equivariant quantum cohomology do not demonstrate simple functoriality, in the sense that a projection of the complete flag variety G/B to a partial flag variety G/P does not induce a homomorphism of rings $QH_T^*(G/P) \rightarrow QH_T^*(G/B)$, and indeed the quantum cohomology of G/P is not isomorphic to the subring of the quantum cohomology of G/B generated by the Schubert classes indexed by W^P . This is reflected in the fact that there are parabolic quantum double Schubert polynomials defined in a different manner that represent Schubert classes in $QH_T^*(G/P)$ [6] and are not in general equal to the quantum double Schubert polynomials above.

Definition 2.5 (Parabolic quantum double Schubert polynomials). Following Lam and Shimozono [6], we define parabolic quantum double Schubert polynomials as follows. Fix an integer $n > 0$. A parabolic subgroup W_P of S_n , which has Coxeter generators $s_1, \dots, s_{n-1}$, is generated by a subset $P \subseteq \{s_1, \dots, s_{n-1}\}$. In terms of the subgroup's properties, it is most natural to identify P by the reflections that it *excludes*. If P is generated by $S \setminus \{s_{i_1}, \dots, s_{i_k}\}$ in increasing order, we define integers $n_1, \dots, n_{k+1}$ with

$$n_1 + \dots + n_{k+1} = n$$

by $n_1 = i_1$ and $n_p = i_p - i_{p-1}$ for $2 \leq p \leq k+1$. Then N_j for $j \geq 1$ is defined by

$$N_j = n_1 + \dots + n_j$$

for $0 \leq j \leq k+1$.

For such a setup, define a matrix D_j^P to be the $n \times n$ matrix with $x_1, \dots, x_n$ on the diagonal, -1 on the superdiagonal, and $(-1)^{n_{j+1}+1}q_j$ at $(N_{j+1}, N_{j-1} + 1)$ for $1 \leq j \leq k$. Notice that there are precisely k q variables, and we set

$$\deg q_j = n_j + n_{j+1}$$

Define

$$\Delta_j^P(t; x) = \det(D_j^P - t\text{Id})$$

Then we define the parabolic quantum double Schubert polynomials $\tilde{\mathfrak{S}}_w^P(x; y)$ for $w \in W^P$ by

$$\tilde{\mathfrak{S}}_w^P(x; y) = (-1)^{\ell(w(w_0^P)^{-1})} \partial_{w(w_0^P)^{-1}}^{(y)} \prod_{j=1}^k \Delta_j^P(-y_{n-N_{j+1}+1}, \dots, -y_{n-N_j}; x)$$

where w_0^P is the longest element of W^P .

Taken as defined, when we vary n these polynomials are not stable under inclusion and do not form a basis of $\mathbb{Z}[q][x_1, \dots, x_n; y]$ as a module over $\mathbb{Z}[q][y]$. To obtain stability and a basis, instead we fix n and choose $N > n$ as large as we would like, maintaining $N_{k+1} = n$ and setting $n_{k+2} = N - n$. From a cohomological perspective, this introduces “extraneous” polynomials that are indexed by elements not in W^P . The full indexing set is $W^{\hat{P}} \subset S_\infty$, where

$$\hat{P} = P \cup \{s_{n+1}, s_{n+2}, \dots\}$$

Example 2.6. As equivariant cohomology rings of finite dimensional projective varieties over $\mathbb{C}$ are finitely generated modules over the coefficient ring, it is fully expected that products of double Schubert polynomials may include terms indexed by permutations that do not lie within the indexing set of the basis of Schubert classes. For example, consider the following non-quantum double Schubert polynomial product, with $P = \{s_1, s_3\}$ and basis elements in S_4 :

$$\begin{aligned} \mathfrak{S}_{2413}(x; y) \mathfrak{S}_{3412}(x; y) = & -(\alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 + \alpha_3)(\alpha_2 + \alpha_3) \mathfrak{S}_{3412}(x; y) \\ & + (\alpha_1 + \alpha_2)(\alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4) \mathfrak{S}_{35124}(x; y) \\ & - (\alpha_1 + \alpha_2) \mathfrak{S}_{361245}(x; y) \\ & - (\alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4) \mathfrak{S}_{45123}(x; y) \\ & + \mathfrak{S}_{461235}(x; y). \end{aligned}$$

Not only does the product $\tilde{\mathfrak{S}}_{2413}^P(x; y) \tilde{\mathfrak{S}}_{3412}^P(x; y)$ include extra terms indexed by permutations in W^P , there are also terms with nonzero coefficients that are in $W^{\hat{P}} \setminus W^P$:

$$\begin{aligned} q_1(\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3 + \alpha_2^2) \tilde{\mathfrak{S}}_{132}^P(x; y) + q_1(\alpha_1 + \alpha_2) \tilde{\mathfrak{S}}_{231}^P(x; y) \\ + q_1(\alpha_2 + \alpha_3) \tilde{\mathfrak{S}}_{1423}^P(x; y) \\ + q_1(\alpha_1 + \alpha_2) \tilde{\mathfrak{S}}_{13254}^{\hat{P}}(x; y) \\ + q_1 \tilde{\mathfrak{S}}_{14253}^{\hat{P}}(x; y) \\ + q_1 \tilde{\mathfrak{S}}_{15234}^P(x; y) \\ + q_1 \tilde{\mathfrak{S}}_{2413}^P(x; y) \end{aligned}$$

Definition 2.7 (q -vectors and Peterson-Woodward lifts). Define $Q^\vee$, for legacy purposes, to be the set of weak compositions. For an integer i and $d \in Q^\vee$, define

$$\langle d \rangle_i = 2d_i - d_{i-1} - d_{i+1}$$

The double indexed version will be denoted by

$$\langle d \rangle_{ij} = \sum_{k=i}^{j-1} \langle d \rangle_k$$

For a parabolic subgroup W_P , we define $Q_P^\vee$ to be the set of weak compositions d such that $\langle d \rangle_i = 0$ for all $s_i \notin P$. The q -vectors in W^P are indexed by $Q^\vee / Q_P^\vee$.

Let $d_P \in Q^\vee / Q_P^\vee$. A *Peterson-Woodward lift* of d_P is a $d \in Q^\vee$ such that $d_P \in d + Q_P^\vee$ and $\langle d \rangle_{ij} \in \{0, -1\}$ for all i, j such that $s_i, \dots, s_{j-1} \in P$.

Method 2.8 (Checking the Peterson-Woodward lift property). While backing out a Peterson-Woodward lift is difficult, going in the other direction is not. Suppose $d \in Q^\vee$ and P is parabolic. If there is any $i \in P$ such that $\langle d \rangle_i \notin \{0, -1\}$, then d is not a Peterson-Woodward lift. If this condition does hold, let $z_P(d)$ be the set of all indexes i such that $\langle d \rangle_i = 0$. This set is naturally divided into maximal disjoint intervals $I_1, \dots, I_m$. We must check the additional condition that for each such I and each pair $i < j$ with $i, j-1 \in I$, the condition $\langle d \rangle_{ij} \in \{0, -1\}$ holds. If and only if it does, then d is a Peterson-Woodward lift of a unique $d_P \in Q^\vee / Q_P^\vee$. We denote this d_P by $\pi_P(d)$.

This algorithm is polynomial in the length of d , which in almost all currently feasible cases is quite small. In contrast, a naive search for d with d_P as the input has an exponential search space, though a more refined algorithm may exist.

We prove the following result.

Proposition 2.9. Let W_P be a parabolic subgroup of S_∞ with $P = \hat{P}$ and let $u, v, w \in W^P$. Then for any $d_P = \pi_P(d) \in Q^\vee / Q_P^\vee$, we have that

$$c_{u,v}^{w, d_P}(y; z) = c_{u,v}^{w w_P w_{z_P(d)}, d}(y; z),$$

Thus, though we lack a simple functoriality property, we can reduce the computation of parabolic triple quantum Schubert structure constants completely to the complete flag variety case with an efficient algorithm. We obtain a nontrivial corollary from this.

Example 2.10. The product in the example above, with $P = \widehat{P}$ satisfying $P \cap \{s_1, s_2, s_3, s_4, s_5\} = \{s_1, s_4\}$, allows us to compute the triple structure constants for the corresponding parabolic quantum double Schubert polynomials:

$$\begin{aligned}
& q_1 q_2 (y_1 + y_2 + y_5 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{1243}^P(x; y) \\
& + q_1 q_2 \tilde{\mathfrak{S}}_{124365}^P(x; y) + q_1 q_2 \tilde{\mathfrak{S}}_{1342}^P(x; y) + q_1 q_2 \tilde{\mathfrak{S}}_{1423}^P(x; y) \\
& + q_2 ((y_1 - z_2)(y_1 - z_3) + (y_1 - z_3)(y_5 - z_1) + (y_5 - z_1)(y_5 - z_2)) \tilde{\mathfrak{S}}_{15234}^P(x; y) \\
& + q_2 (y_1 + y_5 + y_6 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{162345}^P(x; y) + q_2 (y_1 + y_2 + y_5 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{25134}^P(x; y) \\
& + ((y_1 - z_2)(y_1 - z_3)(y_4 - z_2) + (y_1 - z_2)(y_1 - z_3)(y_5 - z_1) + (y_1 - z_2)(y_5 - z_1)(y_5 - z_2) \\
& + (y_1 - z_3)(y_4 - z_1)(y_5 - z_1) + (y_4 - z_1)(y_5 - z_1)(y_5 - z_2)) \tilde{\mathfrak{S}}_{15423}^P(x; y) \\
& + q_2 \tilde{\mathfrak{S}}_{1723456}^P(x; y) + q_2 \tilde{\mathfrak{S}}_{261345}^P(x; y) + q_2 \tilde{\mathfrak{S}}_{35124}^P(x; y) \\
& + ((y_1 - z_2)(y_1 - z_3) + (y_1 - z_3)(y_5 - z_1) + (y_5 - z_1)(y_5 - z_2)) \tilde{\mathfrak{S}}_{156234}^P(x; y) \\
& + ((y_1 - z_2)(y_1 - z_3) + (y_1 - z_2)(y_5 + y_6 - z_1 - z_2) \\
& + (y_1 - z_3)(y_4 - z_1) + (y_4 - z_1)(y_5 + y_6 - z_1 - z_2)) \tilde{\mathfrak{S}}_{164235}^P(x; y) \\
& + ((y_4 - z_1)(y_5 - z_3) + (y_4 - z_1)(y_1 + y_2 - z_1 - z_2) \\
& + (y_5 - z_2)(y_5 - z_3) + (y_5 - z_2)(y_1 + y_2 - z_1 - z_2)) \tilde{\mathfrak{S}}_{25413}^P(x; y) \\
& + ((y_1 - z_1)(y_4 - z_1) + (y_5 - z_2)(y_1 + y_4 - z_1 - z_2)) \tilde{\mathfrak{S}}_{45123}^P(x; y) \\
& + (y_1 + y_5 + y_6 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{165234}^P(x; y) + (y_1 + y_2 + y_5 - z_1 - z_2 - z_3) \tilde{\mathfrak{S}}_{256134}^P(x; y) \\
& + (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{1742356}^P(x; y) + (y_1 + y_4 - z_1 - z_2) \tilde{\mathfrak{S}}_{461235}^P(x; y) \\
& + (y_1 + y_2 + y_4 + y_5 + y_6 - 2z_1 - 2z_2 - z_3) \tilde{\mathfrak{S}}_{264135}^P(x; y) \\
& + (y_4 + y_5 - z_1 - z_2) \tilde{\mathfrak{S}}_{35412}^P(x; y) + (y_4 + y_5 - z_1 - z_2) \tilde{\mathfrak{S}}_{45213}^P(x; y) \\
& + \tilde{\mathfrak{S}}_{1752346}^P(x; y) + \tilde{\mathfrak{S}}_{265134}^P(x; y) + \tilde{\mathfrak{S}}_{2741356}^P(x; y) \\
& + \tilde{\mathfrak{S}}_{356124}^P(x; y) + \tilde{\mathfrak{S}}_{364125}^P(x; y) + \tilde{\mathfrak{S}}_{462135}^P(x; y) \\
& + \tilde{\mathfrak{S}}_{561234}^P(x; y)
\end{aligned}$$

For an illustration, consider $d_P = (-, 0, 1, -, 0)$ for the coefficient of 162345. $q^{d_P} = q_2$. The Peterson-Woodward lift of d_P is $d = (0, 0, 1, 0, 0)$, and $w_P = 21354$. $z_P(d) = \{1\}$, so $w_P w_{z_P(d)} = 12354$. Thus, $ww_P w_{z_P(d)} = 162435$, and the q_2 -component of the coefficient of $\tilde{\mathfrak{S}}_{162345}^P(x; y)$ is $(y_1 + y_5 + y_6 - z_1 - z_2 - z_3)$, which is the same as the q_3 -component (q^d) of the coefficient of $\tilde{\mathfrak{S}}_{162435}(x; y)$ in the complete flag variety case (Example 2.3).

Corollary 2.11 (Triple quantum Schubert positivity for the Grassmannian case). *Suppose $u, v, w \in S_n$ are k -Grassmannian, where P is the corresponding parabolic subgroup such that $n_1 = k$ and $n_2 = n - k$. Then $c_{u,v}^{w;d}(y; z)$ exhibits triple Schubert positivity.*

3. PIERI

Definition 3.1 ($\overset{k}{\rightsquigarrow}$, $\text{fix}_k(u, w)$, $q_k(u, w)$, $E_{p,k}(y; z)$, $\mathbf{E}(y_A; z)$). We define a relation $\overset{k}{\rightsquigarrow}$ on pairs of permutations, introduced in its non-quantum form by Sottile [9], which was used originally to indicate when the

coefficient of a Schubert polynomial is nonzero in a suitable application of the Pieri formula. In the generalization to double Schubert polynomials in [8], the same relation applies to all nonzero coefficients in the expansion of the generalized Pieri formula. The relation plays the same role in our case.

Given $u, u' \in S_\infty$ and a positive integer k , we declare that $u \overset{k}{\rightsquigarrow} w$ if there exists a p with $0 \leq p \leq k$ such that there are transpositions $t_{a_1 b_1}, \dots, t_{a_p b_p}$ satisfying

- (1) $a_i \leq k < b_i$ for all i
- (2) $a_i \neq a_j$ whenever $i \neq j$
- (3) $b_i \geq b_j$ whenever $i < j$
- (4) We have either

$$\ell(ut_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(ut_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) + 1$$

or

$$\ell(ut_{a_1 b_1} \cdots t_{a_i b_i}) = \ell(ut_{a_1 b_1} \cdots t_{a_{i-1} b_{i-1}}) - 2(b_i - a_i) + 1$$

for all i

- (5) $w = ut_{a_1 b_1} \cdots t_{a_p b_p}$

In contrast to the non-quantum case, $u \overset{k}{\rightsquigarrow} w$ does not imply that $\ell(u) \leq \ell(w)$, as the final condition can cause the length of the permutation to drop. For intuition, the condition $\ell(ut_{ab}) = \ell(u) - 2(b - a) + 1$ means that if we write the reflection t_{ab} as

$$s_b s_{b-1} \cdots s_{a+1} s_a s_{a+1} \cdots s_{b-1} s_b$$

then each simple reflection in this word cancels a corresponding simple reflection in a reduced word for u , as opposite to the length-increase case where there is a word for which precisely one simple reflection is inserted, regardless of the length of the reflection as a permutation.

If $u \overset{k}{\rightsquigarrow} w$, define

$$\text{fix}_k(u, w) = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$

Also, we define $q_k(u, w)$ recursively. If $u = w$, define $q_k(u, w) = 1$. If we have defined $q_k(u, w')$ with $w' = t_{a_1 b_1} \cdots t_{a_{p-1} b_{p-1}}$, suppose $\ell(w' t_{a_p b_p}) = \ell(w') + 1$. Then define

$$q_k(u, w) = q_k(u, w')$$

If instead $\ell(w' t_{a_p b_p}) = \ell(w') - 2(b_p - a_p) + 1$, define

$$q_k(u, w) = q_k(u, w') q_{a_p b_p}$$

Suppose y and z are sets of variables and $0 \leq p \leq k$ are integers. We define $E_{p,k}(y; z)$ to be the factorial elementary symmetric polynomial in y and z . For specificity, this is the double Schubert polynomial $\mathfrak{S}_{c[p,k]}(y; z)$ where

$$c[p, k] = s_{k-p+1} \cdots s_{k-1} s_k$$

If $p = 0$, the polynomial is identically 1. If $p = k$, $E_{k,k}(y; z)$ can be simply written as

$$E_{k,k}(y; z) = \prod_{i=1}^k (y_i - z_1)$$

An explicit formula can be written for general p without too much trouble as well, using e.g. [7]:

$$E_{p,k}(y; z) = \sum_{1 \leq i_1 < \cdots < i_p \leq k} \prod_{j=1}^p (y_{i_j} - z_{i_j - j + 1})$$

For a set $A = \{i_1, \dots, i_{|A|}\}$, we define

$$\mathbf{E}(y_A; z) = \prod_{i \in A} (y_i - z_1) = E_{|A|, |A|}(y_{i_1}, \dots, y_{i_{|A|}}; z),$$

the specialization of the factorial elementary polynomial to which the coefficients of the Pieri formula reduce. It will be used to express those coefficients.

Theorem 3.2 (The Pieri formula). *Suppose $u \in S_\infty$ and $k \geq 1$. Then*

$$\tilde{\mathfrak{S}}_u(x; y) \tilde{E}_{k,k}(x; z) = \sum_{u \xrightarrow{k} w} q_k(u, w) \mathbf{E}(y_{\text{fix}_k(u, w)}; z) \tilde{\mathfrak{S}}_w(x; y)$$

4. EFFICIENT MULTIPLICATION OF (QUANTUM) DOUBLE SCHUBERT POLYNOMIALS

Definition 4.1 ($\mathfrak{c}$, (strict) dominant permutations, θ , μ , $\tilde{\theta}$, $\tilde{\mu}$). We say that a permutation μ is *dominant* if $\mathfrak{c}_i(\mu) \geq \mathfrak{c}_{i+1}(\mu)$ for all i . A permutation μ is said to be *strict dominant* if μ is dominant and whenever $\mathfrak{c}_i(\mu) \neq 0$ we have $\mathfrak{c}_i(\mu) > \mathfrak{c}_{i+1}(\mu)$. For a permutation $v \in S_\infty$, we define a dominant permutation $\mu(v)$ recursively as follows. If v is already dominant, define $\mu(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$, and define

$$\mu(v) = \mu(vs_i)$$

Define $\theta_i(v)$ to be $\mathfrak{c}_i(\mu(v))$.

Similarly, for a permutation $v \in S_\infty$, we define a strict dominant permutation $\tilde{\mu}(v)$ recursively as follows. If v is already strict dominant, define $\tilde{\mu}(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_{i+1}(v) \neq 0$ and $\mathfrak{c}_i(v) \leq \mathfrak{c}_{i+1}(v)$, then define

$$\tilde{\mu}(v) = \tilde{\mu}(vs_i)$$

Define $\tilde{\theta}_i(v)$ to be $\mathfrak{c}_i(\tilde{\mu}(v))$.

Definition 4.2 ($\xrightarrow{i}$, $\sigma_i(u, v)$, $D_i(u, v)$). Let $u, v \in S_\infty$ and $i \geq 1$ be an integer. We define $u \xrightarrow{i} v$ if there exist distinct integers $b_1, b_2, \dots, b_k$ such that $b_j \neq i$ for all j , $\ell(ut_{i,b_1} \cdots t_{i,b_k}) = \ell(u) + k$ for all k , and

$$v = ut_{i,b_1} \cdots t_{i,b_k}$$

We define

$$\sigma_i(u, v) = \#\{j \mid b_j < i\}$$

and we define

$$D_i(u, v) = \{u(i)\} \cup \{u(j) \mid u(j) \neq v(j)\}$$

Theorem 4.3 (Formula for ordinary double Schubert polynomials). *Suppose $v \in S_\infty$. Then*

$$\mathfrak{S}_v(x; z) = \sum_{(v_0, \dots, v_k)} \prod_{i=1}^k (-1)^{\sigma_i(v_{i-1}, v_i)} E_{\theta_i(v) - \ell(v_{i-1}, v_i), \theta_i(v)}(x; z_{D_i(v_{i-1}, v_i)})$$

Where $v_{i-1} \xrightarrow{i} v_i$ for all i , $v_0 = 1$, and $v_k = v\mu(v^{-1})$.

Theorem 4.4 (Formula for quantum double Schubert polynomials). *Suppose $v \in S_\infty$. Then*

$$\tilde{\mathfrak{S}}_v(x; z) = \sum_{(v_0, \dots, v_k)} \prod_{i=1}^k (-1)^{\sigma_i(v_{i-1}, v_i)} E_{\tilde{\theta}_i(v) - \ell(v_{i-1}, v_i), \tilde{\theta}_i(v)}^q(x; z_{D_i(v_{i-1}, v_i)})$$

Where $v_{i-1} \xrightarrow{i} v_i$ for all i , $v_0 = 1$, and $v_k = v\tilde{\mu}(v^{-1})$.

Coupled with the Pieri formulas, these formulas give a relatively efficient method for computing products of (quantum) double Schubert polynomials. We describe the resulting algorithm precisely and record its asymptotic complexity in the classical (non-quantum) case. To do so we build it up from its innermost primitive, the elementary (vertical-strip) Pieri step that generates the u -side covers at each layer.

Definition 4.5 (k -strip cover, admissible k -chain, the (p, k) -Pieri problem). Let $w \in S_n$, let $1 \leq k < n$, and let $\beta \in \{1, \dots, n\} \cup \{\infty\}$. A k -strip cover of (w, β) is a transposition t_{ab} of positions a, b with $1 \leq a \leq k < b \leq n$ such that

- (1) $w < wt_{ab}$ in Bruhat order; that is, $w(a) < w(b)$ and no c with $a < c < b$ satisfies $w(a) < w(c) < w(b)$; and
- (2) $w(b) < \beta$.

Its *output* is the pair $(wt_{ab}, w(b))$. An *admissible k -chain of length p* from u is a sequence $u = w_0, w_1, \dots, w_p$ carrying bounds $\infty = \beta_0 > \beta_1 > \dots > \beta_p$ in which (w_t, β_t) is the output of a k -strip cover of (w_{t-1}, β_{t-1}) for every t . The (p, k) -Pieri problem for u asks to enumerate every admissible k -chain from u of length at most p . These chains index the permutations occurring in $e_p(x_1, \dots, x_k) \mathfrak{S}_u$, where e_p is the elementary symmetric polynomial.

Method 4.6 (**piri** (u, p, k)). Given $u \in S_n$, $1 \leq k < n$, and $0 \leq p \leq k$, the algorithm returns a list T of pairs (w, t) , one for each admissible k -chain from u of length $t \leq p$, recording its endpoint w .

- (1) Initialize $T \leftarrow [(u, 0)]$ and a frontier $F \leftarrow [(u, \infty)]$.
- (2) For $t = 1, \dots, p$, form a new frontier $F' \leftarrow []$ and, for each $(w, \beta) \in F$:
 - (a) compute $A \leftarrow \{a \in \{1, \dots, k\} \mid w(a) < \beta\}$;
 - (b) for each $b = k+1, \dots, n$ with $w(b) < \beta$, and each $a \in A$ for which t_{ab} is a k -strip cover of (w, β) , append $(wt_{ab}, w(b))$ to F' and (wt_{ab}, t) to T .
 Then set $F \leftarrow F'$.
- (3) Return T .

Lemma 4.7 (Multiplicity-freeness). *For each t , distinct admissible k -chains from u of length t have distinct endpoints; equivalently, the endpoint map is a bijection from admissible k -chains of length t onto the permutations w with $w \succ^k u$ of length $\ell(u) + t$ produced by the Pieri rule for $e_t(x_1, \dots, x_k)$.*

Proof. The strictly decreasing bounds $\beta_0 > \beta_1 > \dots$ force the values moved leftward across position k to be strictly decreasing, which is exactly the condition making the e_t -Pieri expansion multiplicity-free; see [1]. Reconstructing the chain from its endpoint by peeling off these moves in decreasing order of value shows the map is injective, hence bijective onto the Pieri support. $\square$

Proposition 4.8 (Complexity of **piri**). *Let $M := |T|$ be the number of admissible k -chains from u of length at most p that **piri** (u, p, k) returns. The algorithm runs in*

$$O(k(n-k)n \cdot M)$$

elementary operations, that is $O(n^3)$ per enumerated permutation. Moreover, by Lemma 4.7,

$$M \leq \#\{w \in S_n \mid w \geq u, \ell(w) \leq \ell(u) + p\}.$$

Proof. Processing one frontier node (w, β) costs $O(k)$ to build A , and then iterates over the $n - k$ choices of b and the at most k elements of A ; for each pair the Bruhat-cover test and the transposition each scan $O(n)$ positions. Thus a node costs $O(k + (n - k)kn) = O(k(n - k)n)$. By Lemma 4.7 the number of nodes ever placed on a frontier equals $\sum_{t=0}^{p-1} N_t$, where N_t is the number of length- t chains; since $\sum_{t=0}^{p-1} N_t \leq \sum_{t=0}^p N_t = M$, the total cost is $O(k(n - k)n \cdot M)$. The bound on M holds because every endpoint satisfies $w \geq u$ and $\ell(w) \leq \ell(u) + p$. $\square$

We now describe the v -side primitive. Its starting point is the structure of a *dominant* double Schubert polynomial. If μ is dominant with $\mathbf{c}(\mu) = (\lambda_1, \dots, \lambda_m)$, then $\mathfrak{S}_\mu(x; z)$ factors as a product of factorial elementary symmetric polynomials of maximal degree, one per column of the shape,

$$\mathfrak{S}_\mu(x; z) = \prod_{i=1}^m E_{\lambda_i, \lambda_i}(x; z^{(i)}),$$

where $z^{(i)}$ is the shifted z -alphabet attached to column i (Definition 3.1). For a general v one writes $\mathfrak{S}_v(x; z)$ by applying divided differences in the z -variables to the dominant polynomial $\mathfrak{S}_{\mu(v-1)}(x; z)$; because that polynomial factors by column, the divided-difference action decouples and may be carried out one column (one $z^{(i)}$ -group) at a time. Applied to a single column this produces a *signed* descending expansion. The following primitive carries it out.

Definition 4.9 (k -strip descent beneath an edge, the down-Pieri problem). Let $w, \mu \in S_n$ and $1 \leq k < n$. Say w' lies *beneath the edge* μ if

$$\text{inv}(w'\mu) = \text{inv}(\mu) - \text{inv}(w'),$$

that is, $\ell(w'\mu) = \ell(\mu) - \ell(w')$. A k -strip descent of w is a Bruhat descent $w \succ wt_{ab}$ realized by a transposition t_{ab} that moves the value in the pivot position k either leftward (from a position $a < k$, with sign -1) or rightward (to a position $b > k$, with sign $+1$), touching only positions still holding their original value. Given

a starting label w , the edge μ , a degree p , and the column k , the *down-Pieri problem* asks for the signed list of those length- t descending k -strip chains from w ($t \leq p$) whose endpoint lies beneath μ .

Method 4.10 ($\mathbf{kdown}(w, \mu, p, k)$). The algorithm returns a signed list D of triples (w', t, s) , one for each length- t descending k -strip chain from w ($t \leq p$) whose endpoint w' lies beneath μ , with $s \in \{+1, -1\}$ its sign.

- (1) Initialize $D \leftarrow []$; if w is already beneath μ , append $(w, 0, +1)$. Set a frontier $F \leftarrow [(w, +1)]$.
- (2) For $t = 1, \dots, p$, form $F' \leftarrow []$ and, for each $(w'', s) \in F$:
 - (a) (leftward) for each position $a < k$ with $w''(a) = w(a)$ such that $w'' \succ w''t_{a,k}$ is a Bruhat descent, set $w''' \leftarrow w''t_{a,k}$ and append $(w''', -s)$ to F' ; if w''' is beneath μ , append $(w''', t, -s)$ to D ;
 - (b) (rightward) for each position $b > k$ with $w''(b) = w(b)$ such that $w'' \succ w''t_{k,b}$ is a Bruhat descent, set $w''' \leftarrow w''t_{k,b}$ and append $(w''', +s)$ to F' ; if w''' is beneath μ , append $(w''', t, +s)$ to D .
 Then set $F \leftarrow F'$.
- (3) Return D .

Applied without the edge μ , the signed descending expansion produces many extraneous terms that ultimately cancel. The edge is exactly the bound that discards them: only endpoints beneath μ are retained in D . In the assembled v -side (Method 4.11 below) only these retained labels seed the next column, so the edge prevents the number of live paths from blowing up across columns.

Method 4.11 ($\mathbf{vpaths}(v)$, the v -side path dictionaries). Let $\theta := \theta(v^{-1})$ with nonzero parts in columns $1, \dots, L$, and let $vmu := v\mu(v^{-1})$. Processing columns from $i = L$ down to $i = 1$, maintain for each column a set of live labels (initialized at column L to the single label vmu). At column i , with $k = i$ and edge $\mu_i := \mu$ of the shape truncated below column i , call $\mathbf{kdown}(w, \mu_i, \theta_i, k)$ for each live label w ; record its signed output triples as the transitions out of w , and let their endpoints be the live labels seeding column $i - 1$. The collected transitions form the path dictionaries used by **schubmult**.

Lemma 4.12 (Correctness of the edge filter). *A permutation w' produced by a descending k -strip chain from w is retained by $\mathbf{kdown}(w, \mu, p, k)$ if and only if w' lies beneath μ , i.e. $\text{inv}(w'\mu) = \text{inv}(\mu) - \text{inv}(w')$. In the assembly of Method 4.11 the retained signed labels are exactly the terms of $\mathfrak{S}_v(x; z)$ contributed through column i , so no extraneous (cancelling) label propagates to column $i - 1$.*

Proof. The retention test is the identity $\text{inv}(w'\mu) = \text{inv}(\mu) - \text{inv}(w')$ checked in step (2); this is precisely the condition that $w' \leq \mu$ with $\ell(w'\mu) = \ell(\mu) - \ell(w')$. The factorization of the dominant polynomial by column, together with the column-wise decoupling of the z -divided differences, identifies the beneath- μ endpoints at column i with the surviving monomials of $\mathfrak{S}_v(x; z)$ after processing that column; the extraneous chains are exactly those leaving the shape, which the edge discards before they seed column $i - 1$. $\square$

Proposition 4.13 (Complexity of $\mathbf{kdown}$). *Let $R := |D|$ be the number of retained signed triples and F^* the total number of frontier nodes generated by $\mathbf{kdown}(w, \mu, p, k)$. The algorithm runs in*

$$O(n^2 \cdot F^*)$$

elementary operations, and $R \leq F^$. Consequently $\mathbf{vpaths}(v)$, which invokes $\mathbf{kdown}$ once per live label across the $L \leq \min(n - 1, \ell_v)$ columns and by Lemma 4.12 propagates only retained labels, runs in $O(n^2 \cdot \Phi_v)$ operations, where Φ_v is the total number of frontier nodes generated over all columns.*

Proof. Processing one frontier node scans $O(n)$ candidate positions, each requiring an $O(n)$ Bruhat-descent test and $O(n)$ transposition, so $O(n^2)$ per node; summing over the F^* nodes gives $O(n^2 F^*)$. Each retained triple arises from a distinct frontier node, so $R \leq F^*$. In $\mathbf{vpaths}$, by Lemma 4.12 the labels seeding column $i - 1$ are only the retained (beneath-edge) endpoints of column i , so the live-label count never includes extraneous paths; summing the per-column frontier work gives $O(n^2 \Phi_v)$. $\square$

Method 4.14 (The algorithm **schubmult**). Fix n and let $u, v \in S_n$. The algorithm **schubmult**(u, v) computes the expansion of $\mathfrak{S}_u(x; z) \mathfrak{S}_v(x; z)$ in the Schubert basis as follows. Set $\theta := \theta(v^{-1})$ and let

$$L := \#\{i \mid \theta_i \neq 0\}$$

be the number of nonzero parts. The algorithm proceeds in L layers. It maintains a *front*, a collection of nodes each consisting of an intermediate permutation on the u -side paired with a label on the v -side, initialized to the single node (u, e) . At layer i it replaces each node by the nodes obtained by combining one

u -side cover, generated by $\mathbf{pieri}(\cdot, \theta_i, \theta_i)$ (Method 4.6), with one v -side transition, generated by $\mathbf{vpaths}(v)$ (Method 4.11), accumulating coefficients in the coefficient ring. After all L layers the surviving coefficients at v -label $v\mu(v^{-1})$ give the structure constants.

Definition 4.15 (ℓ_u, ℓ_v, P_u). For $u, v \in S_n$ write $\ell_u := \ell(u)$ and $\ell_v := \ell(v)$, and set

$$P_u := \#\{w \in S_n \mid w \geq u \text{ and } \ell(w) \leq \ell_u + \ell_v\}.$$

Theorem 4.16 (Complexity of $\mathbf{schubmult}$). *The asymptotic complexity of $\mathbf{schubmult}(u, v)$ is*

$$O(\min(n-1, \ell_v) \cdot P_u \cdot 2^{\ell_v} \cdot \text{poly}(n))$$

operations in the coefficient ring, which is $O(2^{\binom{n}{2} + n \log_2 n})$ in the worst case, and $O(2^{n \log_2 n})$ when v is dominant.

Remark 4.17. Note that this does not exceed the time complexity of multiplication of two arbitrary polynomials in n variables in the monomial basis of degree at most $\binom{n}{2}$, which is $O(2^{\binom{n}{2}})$.

Proof. By Method 4.14 the algorithm runs in $L = \#\{i \mid \theta_i(v^{-1}) \neq 0\} \leq \min(n-1, \ell_v)$ layers. Write $\mu := \mu(v^{-1})$, so that $\sum_i \theta_i(v^{-1}) = \ell(\mu)$; note that $\ell(\mu) \geq \ell_v$, with equality if and only if v is dominant, so the total degree processed on the shape is in general strictly larger than ℓ_v . The key point is that the two sides are yoked: at layer i a u -side degree u_i and a v -side degree v_i are paired with $u_i + v_i = \theta_i$, and the v -side descends from $v\mu$ to the identity, so $\sum_i v_i = \ell(v\mu)$. Hence the total degree added on the u -side is

$$\sum_i u_i = \sum_i \theta_i - \sum_i v_i = \ell(\mu) - \ell(v\mu) = \ell_v,$$

using $\ell(v\mu) = \ell(\mu) - \ell(v)$ (as $v^{-1} \leq \mu$ in weak order). Thus every intermediate u -side permutation satisfies $w \geq u$ and $\ell(w) \leq \ell_u + \ell_v$, so there are at most P_u of them. The v -side labels lie in the lower interval $[e, v\mu]$; since $\ell(v\mu) = \ell(\mu)$ can exceed ℓ_v , we bound their number by $2^{\ell(\mu)}$, but the edge filter of Lemma 4.12 retains only those beneath the running dominant edge, of which there are at most 2^{ℓ_v} live at once. Each layer performs work bounded by the product of the front size, the number of live v -labels, and the two per-step branchings, each of which is $\text{poly}(n)$; multiplying by L layers gives the stated bound. Substituting $P_u \leq n! \leq 2^{n \log_2 n}$ and $\ell_v \leq \binom{n}{2}$ yields the worst-case form. If v is dominant then $\theta(v) = \mathbf{c}(v)$, $\ell(\mu) = \ell_v$, every path step carries a $+1$ sign, and each layer's v -side is a single path, so the factor 2^{ℓ_v} is replaced by $O(\ell_v)$ and no term is produced and later canceled, giving $O(\ell_v \cdot P_u \cdot \text{poly}(n)) = O(2^{n \log_2 n})$. $\square$

Remark 4.18. The quantum algorithm has the same shape, running the transition of Theorem 4.4 in place of Theorem 4.3, with the quantum factorial-elementary Pieri operator at each layer. Two things change relative to the classical case. First, the v -side is unaffected: Theorem 4.4 uses the same increasing-path structure (via $\tilde{\mu}$ and $\tilde{\theta}$), so the number of v -labels is still at most 2^{ℓ_v} . Second, on the u -side the quantum covers of Remark 2.1 may *decrease* length, so a node is now a pair (w, d) of a permutation together with a q -degree, and the same w may be reached at several q -degrees. This does not reintroduce cycles: each length-decreasing cover raises the q -degree $\sum_i d_i$ by a strictly positive amount (by $b - a$ when the transposition is t_{ab}), so grading nodes by q -degree—equivalently, by the quantum k -Bruhat order—keeps the transition graph acyclic.

Definition 4.19 ($\hat{P}_u, Q_{u,v}$). For $u, v \in S_n$ set

$$\hat{P}_u := \#\{w \in S_n \mid \ell(w) \leq \ell_u + \ell_v\}, \quad Q_{u,v} := \binom{\lfloor (\ell_u + \ell_v)/2 \rfloor + n - 1}{n - 1}.$$

Theorem 4.20 (Complexity of quantum $\mathbf{schubmult}$). *The asymptotic complexity of the quantum form of $\mathbf{schubmult}(u, v)$ is*

$$O(\min(n-1, \ell_v) \cdot \hat{P}_u \cdot Q_{u,v} \cdot 2^{\ell_v} \cdot \text{poly}(n))$$

operations in the coefficient ring. In the worst case this is at most

$$2^{\binom{n}{2}} \cdot n! \cdot Q_{u,v} \cdot \text{poly}(n),$$

whose leading exponent is $\binom{n}{2}$, the same as in the classical bound of Theorem 4.16.

Proof. The layer count $L \leq \min(n-1, \ell_v)$ and the v -side bound 2^{ℓ_v} are as in Theorem 4.16, since the v -side transition is unchanged. On the u -side, the pairing $\sum_i u_i = \ell(\mu) - \ell(v\mu) = \ell_v$ of Theorem 4.16 still bounds the net length added when all covers are length-increasing; the quantum covers of Remark 2.1 can additionally *decrease* length, which only lowers $\ell(w)$, so every reachable permutation still satisfies $\ell(w) \leq \ell_u + \ell_v$ and there are at most $\widehat{P}_u$ of them. Each such permutation is reached with a q -degree $d = (d_1, \dots, d_{n-1})$; by the grading, a nonzero contribution requires $\ell(u) + \ell(v) - \ell(w) + 2 \sum_i d_i \geq 0$, and specializing $y = z = 0$ (equivalently, comparing with the small quantum cohomology of the flag variety [2]) forces $\sum_i d_i \leq \frac{1}{2}(\ell_u + \ell_v)$ at every layer. The number of such q -degrees is at most $\binom{\lfloor (\ell_u + \ell_v)/2 \rfloor + n - 1}{n-1} = Q_{u,v}$, so the u -front is a multiset of at most $\widehat{P}_u \cdot Q_{u,v}$ visits. Multiplying the per-layer work as before gives the stated bound. For the worst case use $\widehat{P}_u \leq n!$ and $\ell_v \leq \binom{n}{2}$; since $\log_2(n! Q_{u,v}) = O(n \log n) = o\left(\binom{n}{2}\right)$, the leading exponent is $\binom{n}{2}$. $\square$

Definition 4.21 ($\chi_{j;n}, \chi^{(n)}$). Recall the definition of the commuting difference operators $\chi_{j;n}$, with $j \leq n$, defined by Fomin, Gelfand, and Postnikov, as follows:

$$\chi_{j,n} = x_j - \sum_{1 \leq i < j} q_{ij} \partial^{t_{ij}} + \sum_{j < k \leq n} q_{jk} \partial^{t_{jk}}$$

We denote the sequence $(\chi_{1;n}, \dots, \chi_{n;n})$ with n fixed by $\chi^{(n)}$. Recall (CITATION) that if $\mathfrak{S}_v(x)$ has at most n variables, then

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)})(1) = \mathfrak{S}_v(x)$$

and hence

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)(1) = \mathfrak{S}_v(x; z)$$

We specify a more precise formula, as follows.

Theorem 4.22. *Suppose $v \in S_n$. Then*

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z) = \sum_{u \in S_{n,d}} q^d c_{u,v}^{1,d}(x; z) \partial^u$$

Proof. Note that if $\mathfrak{S}_u(x)$ has at most n variables, then

$$\widetilde{\mathfrak{S}}_u(\chi^{(n)}; y) \widetilde{\mathfrak{S}}_v(\chi^{(n)}; z) = \sum_{w,d} q^d c_{u,v}^{w,d}(y; z) \widetilde{\mathfrak{S}}_w(\chi^{(n)}; y)$$

We thus have that

$$(\widetilde{\mathfrak{S}}_u(\chi^{(n)}; y) \widetilde{\mathfrak{S}}_v(\chi^{(n)}; z))(1) = \sum_{w,d} q^d c_{u,v}^{w,d}(y; z) \mathfrak{S}_w(x; y)$$

This is also equal to

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)(\mathfrak{S}_u(x; y))$$

due to the fact that $(\widetilde{\mathfrak{S}}_u(\chi^{(n)}; y))(1) = \mathfrak{S}_u(x; y)$. To obtain the coefficient of ∂^u in $\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)$, we set $y = x$ after applying this operator. Since $\mathfrak{S}_w(x; x) = 0$ unless $w = 1$, we have that

$$\widetilde{\mathfrak{S}}_v(\chi^{(n)}; z)(\mathfrak{S}_u(x; y))|_{y=x} = \sum_d q^d c_{u,v}^{1,d}(x; z)$$

and the result follows. $\square$

Theorem 4.23. *Let k be an integer and let $v, w \in S_\infty$. Then*

$$\sum_d q^d c_{s_k, v}^{w,d}(y; z) = c_{s_k, v}^w(y; z) + \sum_{i \leq k < j} q_{ij} \mathfrak{S}_{vt_{ij}w^{-1}}(y; z)$$

Proof. We have that the desired value is equal to the coefficient of $\mathfrak{S}_w(x; y)$ in

$$\widetilde{\mathfrak{S}}_{s_k}(\chi^{(n)}; y)(\mathfrak{S}_v(x; z))$$

We have that

$$\widetilde{\mathfrak{S}}_{s_k}(\chi^{(n)}; y) = \mathfrak{S}_{s_k}(x; y) + \sum_{i \leq k < j \leq n} q_{ij} \partial^{t_{ij}}$$

Applying this operator, we obtain

$$\sum_{i \leq k < j \leq n} q_{ij} \mathfrak{S}_{vt_{ij}}(x; z) + \sum_w c_{s_k, v}^w(y; z) \mathfrak{S}_w(x; y)$$

Applying the Cauchy formula, this is equal to

$$\sum_{i \leq k < j, u} q_{ij} \mathfrak{S}_{vt_{ij}u^{-1}}(y; z) \mathfrak{S}_u(x; y) + \sum_w c_{s_k, v}^w(y; z) \mathfrak{S}_w(x; y)$$

Setting $u = w$ and allowing n to be sufficiently large, we obtain the result. $\square$

5. QUANTUM TO CLASSICAL REDUCTION FOR GRASSMANNIAN PRODUCTS

Theorem 5.1 ([4, Theorem 2.7]). *We have the following.*

- (1) $c_{u, v}^{w, d}(y; y) = 0$ unless $\text{sgn}_i(w) + \langle d \rangle_i \leq \text{sgn}_i(u) + \text{sgn}_i(v)$ for all i .
- (2) If $\text{sgn}_k(w) + \langle d \rangle_k = \text{sgn}_k(u) + \text{sgn}_k(v) = 2$ for some k , then

$$c_{u, v}^{w, d}(y; y) = c_{us_k, vs_k}^{ws_k, d - \delta_k}(y; y) = \begin{cases} c_{us_k, v}^{ws_k, d - \delta_k}(y; y) & \text{if } \text{sgn}_k(w) = 0 \\ c_{us_k, v}^{ws_k, d}(y; y) & \text{if } \text{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u and v .

Corollary 5.2 ([4, Corollary 2.8]). *Let $u, v, w \in S_\infty$ and $d \in Q^\vee$. Fix an index i .*

- (1) If $\langle d \rangle_i = \text{sgn}_i(u) = 0$ and $\text{sgn}_i(w) = \text{sgn}_i(v) = 1$, then

$$c_{u, v}^{w, d}(y; y) = c_{u, vs_i}^{ws_i, d}(y; y).$$

- (2) If $\langle d \rangle_i = \text{sgn}_i(u) = 1$ and $\text{sgn}_i(w) = \text{sgn}_i(v) = 0$, then

$$c_{u, v}^{w, d}(y; y) = c_{us_i, v}^{ws_i, d - \delta_i}(y; y).$$

This is symmetric in u and v .

Lemma 5.3. *We have that*

$$c_{u, v}^{w, d}(y; z) = \sum_{v_1 \cdot v_2 = v} c_{u, v_2}^{w, d}(y; y) \mathfrak{S}_{v_1}(y; z)$$

Theorem 5.4 (Quantum reduction holds for triple quantum Schubert coefficients). *We have the following.*

- (1) $c_{u, v}^{w, d}(y; z) = 0$ unless $\text{sgn}_i(w) + \langle d \rangle_i \leq \text{sgn}_i(u) + \text{sgn}_i(v)$ for all i .
- (2) If $\text{sgn}_k(w) + \langle d \rangle_k = \text{sgn}_k(u) + \text{sgn}_k(v) = 2$ for some k , then

$$c_{u, v}^{w, d}(y; z) = c_{us_k, vs_k}^{ws_k, d - \delta_k}(y; z) = \begin{cases} c_{us_k, v}^{ws_k, d - \delta_k}(y; z) & \text{if } \text{sgn}_k(w) = 0 \\ c_{us_k, v}^{ws_k, d}(y; z) & \text{if } \text{sgn}_k(w) = 1 \end{cases}$$

This is symmetric in u and v .

Proof. For (1), we note that since $\text{sgn}_i(v_2) \leq \text{sgn}_i(v)$ for any factorization $v_1 \cdot v_2 = v$, we have that $\text{sgn}_i(w) + \langle d \rangle_i \leq \text{sgn}_i(u) + \text{sgn}_i(v_2)$ is a necessary condition for $c_{u, v_2}^{w, d}(y; y) \neq 0$. The result then follows from Lemma 5.3.

For (2), we note that if $\text{sgn}_k(w) + \langle d \rangle_k = \text{sgn}_k(u) + \text{sgn}_k(v) = 2$, then since $\text{sgn}_k(v_2) = \text{sgn}_k(v)$ for any factorization $v_1 \cdot v_2 = v$ such that $c_{u, v_2}^{w, d}(y; y) \neq 0$, Theorem 5.1(2) holds for each term in the sum in Lemma 5.3, and the result follows. $\square$

Corollary 5.5 (z -version of Corollary 5.2(2)). *Let $u, v, w \in S_\infty$ and $d \in Q^\vee$, and fix an index i with $\text{sgn}_i(v) = 0$. If $\langle d \rangle_i = \text{sgn}_i(u) = 1$ and $\text{sgn}_i(w) = 0$, then*

$$c_{u, v}^{w, d}(y; z) = c_{us_i, v}^{ws_i, d - \delta_i}(y; z).$$

Proof. Since $\text{sgn}_i(v_2) \leq \text{sgn}_i(v) = 0$ for every factorization $v_1 \cdot v_2 = v$, the hypotheses of Corollary 5.2(2) hold for each term $c_{u, v_2}^{w, d}(y; y)$ appearing in Lemma 5.3, so $c_{u, v_2}^{w, d}(y; y) = c_{us_i, v_2}^{ws_i, d - \delta_i}(y; y)$. Resumming via Lemma 5.3 gives the claim. $\square$

Lemma 5.6 (Reduction to a coroot supported at k). *Let $u, v, w \in S_\infty$ and let $d \in Q^\vee$. Suppose v has precisely one descent, which is k . If $c_{u,v}^{w,d}(y; z) \neq 0$, then there exist $u', w' \in S_\infty$ and $d' \in Q^\vee$ with $0 \leq d' \leq d$ such that*

$$c_{u,v}^{w,d}(y; z) = c_{u',v}^{w',d'}(y; z)$$

and $\langle d' \rangle_i \leq 0$ for all $i \neq k$. Moreover, if $d' \neq 0$ then $\langle d' \rangle_k > 0$.

Proof. Suppose $\langle d \rangle_i > 0$ for some $i \neq k$, and choose such an i to be maximal. Because v is k -Grassmannian, $\text{sgn}_i(v) = 0$, so Theorem 5.4(1) forces

$$\text{sgn}_i(w) + \langle d \rangle_i \leq \text{sgn}_i(u) + \text{sgn}_i(v) = \text{sgn}_i(u) \leq 1.$$

Since $\langle d \rangle_i \geq 1$, we conclude $\langle d \rangle_i = 1$, $\text{sgn}_i(u) = 1$, and $\text{sgn}_i(w) = 0$. These are exactly the hypotheses of Corollary 5.5 (note $\text{sgn}_i(v) = 0$), so

$$c_{u,v}^{w,d}(y; z) = c_{us_i,v}^{ws_i,d-\delta_i}(y; z).$$

Moreover $\langle d \rangle_i = 2d_i - d_{i-1} - d_{i+1} = 1$ with $d_{i\pm 1} \geq 0$ forces $d_i \geq 1$, so $d - \delta_i \in Q^\vee$ with $0 \leq d - \delta_i \leq d$ and $|d - \delta_i| < |d|$. Replacing (u, w, d) by $(us_i, v$ fixed, $ws_i, d - \delta_i)$ and iterating, the quantity $|d|$ strictly decreases at each step, so the process terminates in a triple (u', w', d') with $\langle d' \rangle_i \leq 0$ for all $i \neq k$.

Finally, for any finitely supported $d' \in Q^\vee$ the values $\langle d' \rangle_i = 2d'_i - d'_{i-1} - d'_{i+1}$ telescope to $\sum_i \langle d' \rangle_i = 0$. Hence $\langle d' \rangle_k = -\sum_{i \neq k} \langle d' \rangle_i \geq 0$, with equality if and only if $\langle d' \rangle_i = 0$ for all i , i.e. $d' = 0$. Thus $d' \neq 0$ implies $\langle d' \rangle_k > 0$. $\square$

6. PROOF OF THE PIERI FORMULA

Lemma 6.1. *Let $u \in S_\infty$ and let $i, j \geq 1$. Then*

$$\begin{aligned} (x_i - z_j) \tilde{\mathfrak{S}}_u(x; y) &= (y_{u(i)} - z_j) \tilde{\mathfrak{S}}_u(x; y) + \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u) + 1}} \text{sign}(p - i) \tilde{\mathfrak{S}}_{ut_{ip}}(x; y) \\ &+ \sum_{\substack{p \neq i \\ \ell(ut_{ip}) = \ell(u) - 2|p - i| + 1}} \text{sign}(p - i) q_{ip} \tilde{\mathfrak{S}}_{ut_{ip}}(x; y) \end{aligned}$$

Definition 6.2 (Descents of a cycle, $\text{desc}_u(c)$). Let c be a cycle and suppose $u \in S_\infty$. Write c as

$$(a_1, \dots, a_n)$$

where $a_n > a_i$ for all $i < n$ (this is a unique way to express the cycle since the a_j are distinct). Then for any i with $1 \leq i < n$ we say that a_i is a *descent* of c with respect to u if $u(a_i) > u(a_{i+1})$. We denote the set of all descents of c with respect to u by $\text{desc}_u(c)$.

Lemma 6.3. *Suppose $u, w \in S_\infty$ and let $k > 0$ be an integer. Then $u \overset{k}{\rightsquigarrow} w$ if and only if there exist pairwise disjoint cyclic permutations $c_1, \dots, c_n$ of respective cycle lengths $p_1 + 1, \dots, p_n + 1$ such that*

$$w = uc_1 \cdots c_n$$

where for each i there exist pairwise distinct positive integers $a_{i,1}, \dots, a_{i,p_i}$ and b_i such that we may write the cycle c_i as

$$c_i = (a_{i,p_i}, a_{i,p_i-1}, \dots, a_{i,1}, b_i)$$

with $a_{i,j} \leq k$ for all $1 \leq j \leq p_i$, $b_i > k$, and $b_i > b_j$ if $i < j$, and such that for each m we have

$$\ell(u, uc_1 \cdots c_m) = \sum_{i=1}^m \left(p_i - 2 \sum_{a \in \text{desc}_u(c_i)} (b_i - a) \right)$$

and

$$\ell(u, w) + \deg q_k(u, w) = p_1 + \cdots + p_n$$

Proof. Suppose first that $u \xrightarrow{k} w$. By Remark 2.1 there are transpositions $t_{a_1 b_1}, \dots, t_{a_p b_p}$ with $a_i \leq k < b_i$, the a_i pairwise distinct, and $b_1 \geq \dots \geq b_p$, such that

$$u = u_0 \leq_k^{\mathbf{q}} u_1 \leq_k^{\mathbf{q}} \dots \leq_k^{\mathbf{q}} u_p = w, \quad u_i = u_{i-1} t_{a_i b_i},$$

is a saturated chain in the quantum k -Bruhat order. Grouping the transpositions that share a common top b assembles them, exactly as in the non-quantum case [8, Lemma 7.3], into a cycle

$$c = (a_{i+r}, \dots, a_i, b) = t_{a_i b} \dots t_{a_{i+r} b};$$

ranging over the distinct values of b in decreasing order yields pairwise disjoint cycles $c_1, \dots, c_n$ with $w = uc_1 \dots c_n$ of the asserted form. To compute the length, set $c^{(j)} = t_{a_i b} \dots t_{a_j b} = (a_j, \dots, a_i, b)$; the step from $uc^{(j-1)}$ to $uc^{(j)} = uc^{(j-1)} t_{a_j b}$ exchanges the value $u(a_{j-1})$ in position b with the value $u(a_j)$ in position a_j . By the cover characterization of Remark 2.1 this step is a classical cover, of length increment $+1$, when $u(a_j) < u(a_{j-1})$, and a quantum cover, of length increment $-(2(b - a_j) - 1)$, when $u(a_j) > u(a_{j-1})$, that is, precisely when $a_j \in \text{desc}_u(c)$. Summing these increments over the steps constituting $c_1, \dots, c_m$ gives

$$\ell(u, uc_1 \dots c_m) = \sum_{i=1}^m \left(p_i - 2 \sum_{a \in \text{desc}_u(c_i)} (b_i - a) \right),$$

and, recording each quantum step in the q -degree as in Definition 3.1, $\ell(u, w) + \deg q_k(u, w) = p_1 + \dots + p_n$.

Conversely, suppose pairwise disjoint cycles $c_1, \dots, c_n$ as in the statement are given, and write $c_i = t_{a_{i,1} b_i} \dots t_{a_{i,p_i} b_i}$. Applying these transpositions in order, the step from $uc^{(j-1)}$ to $uc^{(j)} = uc^{(j-1)} t_{a_{i,j} b_i}$ exchanges the values $u(a_{i,j})$ and $u(a_{i,j-1})$ in positions $a_{i,j} \leq k < b_i$. Any transposition t_{ab} changes the length by at least $+1$ when it places the larger of the two exchanged values in the smaller position a , and by at least $-(2(b - a) - 1)$ in every case, with equality in either bound if and only if the step is a classical, respectively quantum, cover in the sense of Remark 2.1. Applying the first bound at the non-descent steps and the second at the descents therefore gives

$$\ell(u, uc_1 \dots c_n) \geq p_1 + \dots + p_n - 2 \sum_{i=1}^n \sum_{a \in \text{desc}_u(c_i)} (b_i - a),$$

and the length hypothesis of the statement forces equality throughout, so every step is a cover of the quantum k -Bruhat order and hence $u \xrightarrow{k} w$. $\square$

Definition 6.4 ($\triangleleft$). If $u, u' \in S_\infty$, we define $u \triangleleft u'$ if $\ell(u, u') = 1$ or $u' = ut_{ab}$ where $\ell(u, u') = 1 - 2(b - a)$. Equivalently, $u \triangleleft ut_{ab}$ exactly when $u \leq_k^{\mathbf{q}} u' ut_{ab}$ is a cover of the quantum k -Bruhat order for some (equivalently, any) k with $a \leq k < b$, in the sense of Remark 2.1.

Lemma 6.5. Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. Suppose $u \xrightarrow{k} w$ and let $c = (a_p, \dots, a_1, b)$ be a cycle as in Lemma 6.3 in the decomposition of $u^{-1}w$. Then $\text{desc}_u(c)$ has at most one element.

Proof. This is clearly true if $p = 1$, or if the result holds for $p - 1$ but $c^{(p-1)}$ has no descents. Suppose then that $c^{(p-1)}$ has a descent, say a_i for exactly one $i < p$ by the induction hypothesis, and that $u(a_p) > u(a_{p-1})$, meaning that $a_p \in \text{desc}_u(c^{(p-1)})$. a_i being a descent would require that

$$u(a_i) > uc^{(i-1)}(j) > u(a_{i-1})$$

for all j with $a_i < j < b$. Furthermore, we also have that for all j with $a_p < j < b$ that

$$u(a_p) > uc^{(p-1)}(j) > u(a_{p-1})$$

Suppose $a_p < a_i$. Then $u(a_p) > uc^{(p-1)}(a_i) = u(a_{i-1}) > u(a_{p-1})$. This means that necessarily $a_{p-1} < a_i$, since $u(a_{i-1}) = uc^{(i)}(a_i)$ is minimal in the range $a_i \leq j \leq b$ in $uc^{(i)}$. For the same reason, we cannot have that $uc^{(p-2)}(b) < u(a_{i-1})$, because this would have to come from some $t_{a_q b}$ with $i < q < p - 1$, $a_q < a_i$, and $u(a_q) < u(a_{i-1})$, which would increase the length by more than 1. However, since $uc^{(p-2)}(b) > u(a_{i-1}) > u(a_{p-1})$, this would imply that $t_{a_{p-1} b}$ increases the length by more than 1, which is a contradiction. Thus we cannot have a descent a_p with $a_p < a_i$.

If instead $a_p > a_i$, then $u(a_i) > u(a_p) > u(a_{i-1})$. Since a_p is a descent, $uc^{(p-1)}(b) = u(a_{p-1}) < u(a_p)$, but $uc^{(i)}(b) = u(a_i) > u(a_p)$. This would imply that there is some q with $i < q \leq p - 1$ such that $a_q > a_p$ and

$uc^{(q-1)}t_{a_q b}(b) = u(a_q) < u(a_p)$, because if we had $a_q < a_p$ then the length would increase by more than 1. Let q be minimal with this property. In that case, $u(a_{q-1}) > u(a_p)$ and $uc^{(p-1)}(a_q) = u(a_{q-1})$, contradicting the fact that a_p is a descent.

Having excluded the cases $a_p > a_i$ and $a_p < a_i$, by the assumption that $a_p \neq a_i$ it follows by induction that there can be at most one descent. $\square$

Definition 6.6 (Redefinition of $\text{desc}_u(c)$). If c is a cycle, we redefine $\text{desc}_u(c)$ to be the unique element of the previously defined set $\text{desc}_u(c)$ if it exists, or otherwise 0.

Lemma 6.7. Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. Suppose $u \xrightarrow{k} w$ and let $c = (a_p, \dots, a_1, b)$ be a cycle as in Lemma 6.3 in the decomposition of $u^{-1}w$, with $a_q = k$ being a descent. Then for all $j \geq q$ we have that $u(a_j) > u(b)$.

Proof. We have that $u(a_q) > u(a_{q-1})$. By Lemma 6.5, k is the only descent of c . We claim that for all r with $k \leq r < b$ we have that $u(r) > u(b)$. This is because for all such r we have that $u(a_{q-1}) < u(r)$ since a_q is a descent, and if $u(b) > u(r)$ for some such r then we may pick a maximal $\ell < q$ such that $u(a_\ell) > u(r)$ for some such r . Then $u(a_{\ell+1}) < u(r)$ for all such r , and the reflection $t_{a_{\ell+1}b}$ would increase the length by more than 1, which cannot happen. For the same reason, we cannot have that $u(a_m) < u(r)$ for any r with $k < r \leq b$, since $u(a_q) > u(r)$ for all such r and hence the reflection $t_{a_{m-1}a_m}$ would increase the length by more than q . Thus since in particular $u(r) > u(b)$ for all such r , we must have that $u(a_j) > u(b)$ for all $j \geq q$, as desired. $\square$

Lemma 6.8. Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer. If $u \xrightarrow{k} w$ but $u \not\xrightarrow{k-1} w$, then there is a unique $b > k$ such that $wt_{kb} \triangleleft w$ and $u \xrightarrow{k-1} wt_{kb}$. Furthermore, in that case there does not exist a $a < k$ such that $wt_{ak} \triangleleft w$ and $u \xrightarrow{k-1} wt_{ak}$.

Proof. Suppose $u \xrightarrow{k} w$ but $u \not\xrightarrow{k-1} w$. Let $c_1, \dots, c_n$ be the pairwise disjoint cycles in the decomposition of $u^{-1}w$, with the cycle length of c_i being $p_i + 1$. Some such cycle must contain k , as otherwise we would have $u \xrightarrow{k-1} w$ by Lemma 6.3. Assume that $c_i = (a_{i,p_1}, \dots, a_{i,1}, b_i)$ is the cycle that contains k and set $b = b_i$. We claim that $u \xrightarrow{k-1} wt_{kb}$ and $wt_{kb} \triangleleft w$. Note that $w = (wt_{kb})t_{kb}$ and hence

$$(uc_1 \cdots c_n t_{kb})t_{kb} = w$$

Let j be the index such that $a_j = k$. Then we have

$$\begin{aligned} (a_{i,p_i}, \dots, a_{i,j+1}, k, a_{i,j-1}, \dots, a_{i,1}, b_i)t_{kb} &= (a_{i,j-1}, \dots, a_{i,1}, b, a_{i,p_i}, \dots, a_{i,j+1}, k)t_{kb} \\ &= (a_{i,j-1}, \dots, a_{i,1}, b)(a_{i,p_i}, \dots, a_{i,j+1}, k) \end{aligned}$$

Note that $wt_{kb}(k) = w(b) = u(a_{i,p_i})$ and $wt_{kb}(b) = w(k) = u(a_{i,j-1})$. We need that

$$\ell(u, wt_{kb}) = p'_i + p''_i - 2(b - \text{desc}_u(c'_i)) - 2(k - \text{desc}_u(c''_i)) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

We have that

$$\ell(u, w) = p_i - 2(b - \text{desc}_u(c_i)) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

and

$$\ell(u, wt_{kb}) \geq p'_i + p''_i - 2(b - \text{desc}_u(c'_i)) - 2(k - \text{desc}_u(c''_i)) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

If c'_i has a descent, then c''_i does not, and we are done. If c''_i has a descent, then

$$w(k) = u(a_{i,j-1}) < u(b) < u(a_{i,p_i}) = w(b)$$

by Lemma 6.7, so

$$\ell(w) < \ell(wt_{bk}) \leq \ell(w) + 2(b - k) - 1$$

so

$$\ell(u, wt_{kb}) = \ell(wt_{bk}) - \ell(u) \leq p'_i + p''_i - 2(b - \text{desc}_u(c'_i)) - 2(b - \text{desc}_u(c''_i)) + 2(b - k) + \ell(u, uc_1 \cdots c_{i-1}c_{i+1} \cdots c_m)$$

Since we already have the opposite inequality, we are done if c''_i has a descent, since $wt_{bk} \triangleleft w$. If neither has a descent but c_i does, then this descent must be at k . We have that $wt_{bk}(k) = u(a_{i,p_i}) > u(b)$ by Lemma 6.7, and $wt_{bk}(b) = u(a_{i,j-1}) < u(b)$. Furthermore, $u(k) > u(b) > u(r) > u(a_{i,j-1})$ for all r with $k < r < b$ since k is a descent. Thus $wt_{bk} \triangleleft w$.

$a_{i,q} < r < b$, so the length decreases by $2(b - a_{i,q}) - 1$ as desired. If $u(a_{i,j+1}) > u(k)$

The uniqueness of b among the indices $r > k$ with $\ell(ut_{kr}) = \ell(u) + 1$, and the nonexistence of an index $q < k$ with $\ell(ut_{qk}) = \ell(u) + 1$ and $ut_{qk} \xrightarrow{k-1} w$, now follow exactly as in the proof of [8, Lemma 7.6]. Those arguments use only the disjoint cycle characterization of the Pieri relation, which in the present setting is supplied by Lemma 6.3 in place of [8, Lemma 7.3]: in each case the reflection t_{kr} (respectively t_{qk}) either merges the cycle c_i containing k with another cycle or splits it so that k and a second index exceeding $k - 1$ lie in a common cycle of $(ut_{kr})^{-1}w$ (respectively $(ut_{qk})^{-1}w$), whence $ut_{kr} \xrightarrow{k-1} w$ (respectively $ut_{qk} \xrightarrow{k-1} w$) by Lemma 6.3. $\square$

We organize the inductive step of Theorem 3.2 around the position k , distinguishing whether or not $u(k) = w(k)$. The following lemma records the two combinatorial facts about the interaction of $\xrightarrow{k}$ and $\xrightarrow{k-1}$ at position k that make this dichotomy exhaustive.

Lemma 6.9. *Suppose $u, w \in S_\infty$ and let $k > 1$ be an integer.*

- (i) *If $u(k) = w(k)$, then $u \xrightarrow{k} w$ if and only if $u \xrightarrow{k-1} w$.*
- (ii) *If $u(k) \neq w(k)$ and $u \xrightarrow{k} w$, then $u \xrightarrow{k-1} w$.*

Proof. Both statements follow from the cycle description of Lemma 6.3, according to which $u \xrightarrow{m} w$ if and only if in the disjoint cycle decomposition of $u^{-1}w$ each cycle contains exactly one index exceeding m , subject to the length conditions. If $u(k) = w(k)$ then k is a fixed point of $u^{-1}w$, hence lies in no nontrivial cycle; the constraint distinguishing level k from level $k - 1$ concerns only the indices exceeding $k - 1$ that occur in the cycles, and k is not among them. Thus the cycle conditions for level k and level $k - 1$ coincide, proving (i). For (ii), if $u(k) \neq w(k)$ then k lies in a nontrivial cycle c of $u^{-1}w$; if $u \xrightarrow{k} w$, the level- k condition forces k to be the unique index of c exceeding $k - 1$ that is admissible at level k , but then c contains an index $b > k$ as its cycle top, so c contains two indices $\geq k$, violating the level- $(k - 1)$ condition. Hence $u \xrightarrow{k-1} w$. $\square$

The heart of the induction is the following cancellation, which replaces the informal “the sums are empty” step of the classical argument. When $u(k) = w(k)$ the diagonal term alone already reproduces $c_k(u, w)$, by adjoining the variable $y_{u(k)}$ to the product $\mathbf{E}(y_{\text{fix}_{k-1}}; z)$; the reflection sums and the quantum term must therefore cancel among themselves.

Lemma 6.10 (Reflection-drop cancellation). *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u(k) = w(k)$. Then*

$$\sum_{a < k, wt_{ak} \triangleleft w} \text{sign}(a - k) \kappa_{ak} c_{k-1}(u, wt_{ak}) + \sum_{b > k, wt_{kb} \triangleleft w} \text{sign}(b - k) \kappa_{kb} c_{k-1}(u, wt_{kb}) + q_{k-1} c_{k-2}(u, w) = 0,$$

where $\kappa_{rs} = q_{rs}$ if $wt_{rs} \triangleleft w$ is length-decreasing and $\kappa_{rs} = 1$ if it is length-increasing. In fact the two reflection sums contribute at most one nonzero summand in total: it occurs precisely when $u \xrightarrow{k-2} w$ (equivalently, when the quantum term is nonzero), it is indexed by some $a < k$, and it equals $-q_{k-1} c_{k-2}(u, w)$.

Proof. Throughout we use that $u(k) = w(k)$, so that k is a fixed point of $\pi := u^{-1}w$.

Step 1: no reflection t_{kb} with $b > k$ contributes. Suppose $b > k$ and $wt_{kb} \triangleleft w$ with $c_{k-1}(u, wt_{kb}) \neq 0$, so $u \xrightarrow{k-1} v$ for $v := wt_{kb}$. Then $u^{-1}v = \pi t_{kb}$; since k is a fixed point of π and $b > k$, right multiplication by t_{kb} adjoins k to the π -cycle containing b (or creates the 2-cycle (kb)). Either way the resulting cycle of $u^{-1}v$ contains two indices exceeding $k - 1$, namely k and an index $\geq b > k$, contradicting the level- $(k - 1)$ condition of Lemma 6.3. Hence only reflections t_{ak} with $a < k$ can contribute, and each such summand carries the sign $\text{sign}(a - k) = -1$.

Step 2: the reflection sum has at most one nonzero term, occurring precisely when $u \xrightarrow{k-2} w$. A reflection t_{ak} with $a < k$ contributes precisely when $u \xrightarrow{k-1} v$ for $v := wt_{ak}$; here $u^{-1}v = \pi t_{ak}$, so t_{ak} adjoins the fixed point k to the π -cycle containing a (or creates the 2-cycle (ak) when a is fixed by π). By Lemma 6.3, $u \xrightarrow{k-1} v$ requires this enlarged cycle to contain exactly one index exceeding $k - 1$; as it already contains k , it may contain no further index $\geq k$. Since k is fixed by π , an index occurring in a nontrivial π -cycle and exceeding $k - 1$ is in fact $\geq k + 1$, unless it equals $k - 1$. Thus the enlarged cycle is admissible only if the π -cycle receiving k has all indices $\leq k - 1$, i.e. its top equals $k - 1$; the π -cycles being disjoint, at most one

$a < k$ contributes. Such a configuration exists exactly when $u \overset{k-2}{\rightsquigarrow} w$: if $u \overset{k-2}{\rightsquigarrow} w$ and $k-1$ is not fixed by π , then the π -cycle through $k-1$ has top $k-1$ (any second index $\geq k-1$ would be $\geq k+1$, violating the level- $(k-2)$ condition), while if $k-1$ is fixed the choice $a = k-1$ creates the 2-cycle $(k-1\ k)$; conversely a contributing t_{ak} forces a π -cycle of top $k-1$ (or $k-1$ fixed with $a = k-1$), whence $u \overset{k-2}{\rightsquigarrow} w$. This is exactly the condition under which the quantum term is nonzero.

Step 3: the contributing term cancels the quantum term. Assume $u \overset{k-2}{\rightsquigarrow} w$, let t_{a^*k} ($a^* < k$) be the unique contributing reflection, and set $v := wt_{a^*k}$. By Step 2 the cycle of $u^{-1}v$ containing k has top k . Since v agrees with w off $\{a^*, k\}$ and $v(a^*) = w(k) = u(k)$, both a^* and $k-1$ leave the agreement set in passing to level $k-1$, and comparison of the invariants of Definition 3.1 gives

$$\text{fix}_{k-1}(u, v) = \text{fix}_{k-2}(u, w), \quad n_{k-1}(u, v) = n_{k-2}(u, w) + 1$$

(using $n_m = m - |\text{fix}_m|$). Hence $c_{k-1}(u, v)$ and $c_{k-2}(u, w)$ carry the *same* factorial elementary polynomial $\mathbf{E}(y_P; z)$ with $P = \text{fix}_{k-2}(u, w)$. Finally the q -weights match. Write $\kappa_{a^*k} = 1$ if t_{a^*k} is length-increasing and $\kappa_{a^*k} = q_{a^*k}$ if length-decreasing. As the q -weight is multiplicative over the cycles of $u^{-1}(\cdot)$ and only the cycle receiving k is altered (from top $k-1$ to top k), the definition of the weights yields

$$\kappa_{a^*k} q_{k-1}(u, v) = q_{k-1} q_{k-2}(u, w).$$

Therefore the contributing summand is

$$\text{sign}(a^* - k) \kappa_{a^*k} c_{k-1}(u, v) = -\kappa_{a^*k} q_{k-1}(u, v) \mathbf{E}(y_P; z) = -q_{k-1} q_{k-2}(u, w) \mathbf{E}(y_P; z) = -q_{k-1} c_{k-2}(u, w),$$

cancelling the quantum term. When $u \overset{k-2}{\rightsquigarrow} w$ the quantum term vanishes and, by Step 2, no reflection contributes; all three terms are 0. In every case the displayed sum is 0. $\square$

Remark 6.11. The content of Lemma 6.10 is the identification of the quantum term $q_{k-1}c_{k-2}(u, w)$ with the unique level- $(k-1)$ neighbor v satisfying $u \overset{k-1}{\rightsquigarrow} v$, $u \overset{k-2}{\rightsquigarrow} v$, $\text{fix}_{k-1}(u, v) = \text{fix}_{k-2}(u, w)$ and $n_{k-1}(u, v) = n_{k-2}(u, w) + 1$, obtained by adjoining the fixed point k to the π -cycle of top $k-1$. The connecting reflection t_{a^*k} may be length-increasing or length-decreasing; in either case its weight κ_{a^*k} satisfies the balance $\kappa_{a^*k} q_{k-1}(u, v) = q_{k-1} q_{k-2}(u, w)$. Uniqueness of v is a consequence of the cycle structure (there is a single π -cycle of top $k-1$); this intrinsic matching by the invariants (P, n) is what the $u(k) \neq w(k)$ branch of Lemma 6.13 invokes.

When $u(k) \neq w(k)$ the position k lies in a nontrivial cycle of $\pi = u^{-1}w$, and the reflection sums of the vanishing identity below no longer telescope against the diagonal; instead they cancel among themselves and against the quantum term, one q -weight at a time. The following lemma isolates this cancellation. It is the moved-case counterpart of Lemma 6.10, and its non-quantum instances ($M = 1$) reduce to the pairing of an up-neighbor against a down-neighbor effected by [8, Lemma 7.9] in the cancellation of the classical Pieri rule [8, Proof of Proposition 4.4].

Lemma 6.12 (Moved reflection cancellation). *Let $u, w \in S_\infty$, let $k > 1$, and suppose $u \overset{k}{\rightsquigarrow} w$ with $u(k) \neq w(k)$. Let $\pi = u^{-1}w$, let C be the π -cycle containing k — nontrivial, since $u(k) \neq w(k)$ — and set $B = \max(C)$. Fix a q -monomial $q^M \neq q_{k-1}(u, w)$, and let $\mathcal{G}_M$ be the set of Monk summands of (6.3)–(6.4) of q -weight q^M , augmented by the quantum term $q_{k-1}c_{k-2}(u, w)$ of (6.5) when $q^M = q_{k-1}q_{k-2}(u, w)$. Then the summands of $\mathcal{G}_M$ sum to 0.*

Proof. A neighbor $v = wt$, with $t \in \{t_{ak} : a < k\} \cup \{t_{kb} : b > k\}$, contributes to $\mathcal{G}_M$ only if $u \overset{k-1}{\rightsquigarrow} v$; writing $\sigma = u^{-1}v = \pi t$, Lemma 6.3 requires every cycle of σ to contain exactly one index exceeding $k-1$. Since t moves π only at its two positions, one of which is $k \in C$, the passage $\pi \rightsquigarrow \sigma$ either *splits* C (when the other position lies in C) or *merges* C with the component of the other position (when it does not); every other π -cycle is unchanged and so must already satisfy the level- $(k-1)$ condition.

Up-neighbors are unique and force C to have exactly the indices k, B above $k-1$. Let $v = wt_{kb}$ with $b > k$. If $b \notin C$, the merge produces a cycle containing k, b and $B \geq b$, hence two indices exceeding $k-1$: inadmissible. Thus $b \in C$, and t_{kb} splits C into a piece through k and a piece through b ; admissibility forces each to hold a single index above $k-1$, so b is that index for its piece. As $B = \max(C) > k-1$, this yields $b = B$ and shows C has exactly the two indices k, B above $k-1$, with $B > k$. Hence a contributing up-neighbor, if present, is unique and equals wt_{kB} .

Down-neighbors. Let $v = wt_{ak}$ with $a < k$. If $a \in C$ (a *split*), admissibility assigns the indices of C above $k - 1$ one to each nontrivial piece: if C has three or more such indices no split is admissible; if it has exactly two, k and $B > k$, the split separates them (and then wt_{kB} is the up-neighbor); if it has only k , so $B = k$, the split peels a fixed point off C . If $a \notin C$ (a *merge*), then $C \cup (\text{component of } a)$ must retain a single index above $k - 1$, namely k , forcing $B = k$ and the component of a to have all indices $\leq k - 1$. We record the resulting q -weights.

(1) *The case $B = k$.* Either a is adjoined as a fixed point — then $v(a) = w(k) = u(a)$ reinstates below k the agreement lost at k , the cycle tops are unchanged, and the summand carries weight $q_{k-1}(u, w)$ — or the component absorbed into C is a nontrivial cycle, whose unique index above $k - 2$ is then $k - 1$ (being $\leq k - 1$ and $> k - 2$); the merged cycle acquires the extra factor q_{k-1} , and a direct comparison of the invariants of Definition 3.1 gives

$$\text{fix}_{k-1}(u, v) = \text{fix}_{k-2}(u, w), \quad n_{k-1}(u, v) = n_{k-2}(u, w) + 1, \quad \kappa_{ak}q_{k-1}(u, v) = q_{k-1}q_{k-2}(u, w),$$

the very identities of Lemma 6.10, with the π -cycle of top $k - 1$ now attached to C in place of the fixed point k ; the computation is otherwise verbatim, and by uniqueness of the π -cycle of top $k - 1$ there is at most one such v .

(2) *The case $B > k$.* Here only the split with $C = \{k, B\}$ above $k - 1$ occurs, and it separates k from B .

Cancellation. Suppose first $q^M = q_{k-1}q_{k-2}(u, w)$, so the quantum term lies in $\mathcal{G}_M$ and $u \overset{k-2}{\rightsquigarrow} w$; then every π -cycle has exactly one index above $k - 2$, in particular $B = k$, and by the up-paragraph $\mathcal{G}_M$ contains no up-neighbor. By (1) the summands of weight q^M reduce to the single merge down-neighbor v with $\text{fix}_{k-1}(u, v) = \text{fix}_{k-2}(u, w)$; then $c_{k-1}(u, v)$ and $c_{k-2}(u, w)$ carry the same factorial elementary polynomial $\mathbf{E}(y_{\text{fix}_{k-2}(u, w)}; z)$, and the down summand, of sign $\text{sign}(a - k) = -1$, equals $-q_{k-1}c_{k-2}(u, w)$ and cancels the quantum term. (If no such v exists the quantum term already vanishes.)

Now suppose $q^M \neq q_{k-1}(u, w)$, $q_{k-1}q_{k-2}(u, w)$. By (1) every merge or peel down carries weight $q_{k-1}(u, w)$ or $q_{k-1}q_{k-2}(u, w)$, hence lies outside $\mathcal{G}_M$; so every summand of $\mathcal{G}_M$ is of split type with $C = \{k, B\}$, $B > k$ (whence also $u \overset{k-1}{\rightsquigarrow} w$). For each such summand $\text{fix}_{k-1}(u, v) = \text{fix}_{k-1}(u, w)$: the up wt_{kB} agrees with w below k , while a split down wt_{ak} agrees with w below k except at $a \in C$, where it takes the value $w(k)$; but a is not fixed by π , and $u(a) \neq w(k)$ (otherwise, by (1), the summand would have weight $q_{k-1}(u, w)$), so no agreement below k is created or destroyed. Consequently all summands of $\mathcal{G}_M$ share the factorial elementary factor $E_0 = \mathbf{E}(y_P; z)$, where $P = \text{fix}_{k-1}(u, w)$. Finally, when $C = \{k, B\}$ there are exactly two admissible neighbors through position k : the up wt_{kB} , and — since C has a single descent (Lemma 6.5) and, as in the proof of Lemma 6.8, exactly one covering split of C separates k from B — a single down wt_{a^*k} . The two reflections differ by the 3-cycle (a^*, B, k) ; each resolves the cycle C into two cycles of tops k and B while leaving the other cycles untouched, so by the multiplicativity of the q -weight (Definition 3.1) and Lemma 6.7, which fixes $\text{desc}_u(C)$ and hence the drop factors, both $\kappa_{kB}q_{k-1}(u, wt_{kB})$ and $\kappa_{a^*k}q_{k-1}(u, wt_{a^*k})$ equal the weight q^M of $\mathcal{G}_M$. With opposite signs $\text{sign}(B - k) = +1$ and $\text{sign}(a^* - k) = -1$ and common coefficient E_0 , the up and the down cancel. Hence $\mathcal{G}_M = 0$. $\square$

The remaining case of the induction is when $u \overset{k}{\rightsquigarrow} w$, where the target coefficient is 0.

Lemma 6.13 (Vanishing). *Suppose $u, w \in S_\infty$, let $k > 1$, and suppose $u \overset{k}{\rightsquigarrow} w$. Then*

$$\begin{aligned} (y_{w(k)} - z_1) c_{k-1}(u, w) + \sum_{b > k, wt_{kb} \triangleleft w} \kappa_{kb} c_{k-1}(u, wt_{kb}) - \sum_{a < k, wt_{ak} \triangleleft w} \kappa_{ak} c_{k-1}(u, wt_{ak}) \\ + q_{k-1} c_{k-2}(u, w) = 0, \end{aligned}$$

where $\kappa_{rs} = q_{rs}$ if $wt_{rs} \triangleleft w$ is length-decreasing and $\kappa_{rs} = 1$ if it is length-increasing.

Proof. Suppose first that $u(k) = w(k)$. Then by Lemma 6.9(i) also $u \overset{k-1}{\rightsquigarrow} w$, so $c_{k-1}(u, w)$ vanishes; hence the diagonal term is 0, and the remaining reflection sums and the quantum term cancel by Lemma 6.10. This settles the case $u(k) = w(k)$ completely.

Now suppose $u(k) \neq w(k)$. Since $u \overset{k}{\rightsquigarrow} w$, Lemma 6.3 shows the cycle of $u^{-1}w$ containing k has a top $b > k$, so $c_k(u, w) = 0$; writing F for the right-hand side (6.2)–(6.5), we must show $F = 0$.

We grade by the cohomological degree $\deg q_i = 2$. Under this grading (6.1) is homogeneous, and the q_i are algebraically independent over $\mathbb{Z}[y, z]$. By the definition of c_m , each summand of F is a monomial in

the q_i times a factorial elementary polynomial in $\mathbb{Z}[y, z]$; collecting by q -monomial we write $F = \sum_M q^M F_M$ with $F_M \in \mathbb{Z}[y, z]$, and

$$F = 0 \iff F_M = 0 \text{ for every } q\text{-monomial } M.$$

Fix M and let $\mathcal{G}_M$ be the collection of summands of F whose q -weight equals q^M ; here the diagonal (6.2) carries weight $q_{k-1}(u, w)$, the quantum term carries $q_{k-1}q_{k-2}(u, w)$, and the Monk summand of (6.3)–(6.4) reaching $w' = wt_{ak}$ or $w' = wt_{kb}$ carries $\kappa q_{k-1}(u, w')$ with $\kappa \in \{1, q_{ak}, q_{kb}\}$ its drop-weight. We show $F_M = 0$ in two exhaustive cases.

Case (i): $\mathcal{G}_M$ contains the diagonal. It carries weight $q_{k-1}(u, w)$, so $q^M = q_{k-1}(u, w)$; in particular $u \xrightarrow{k-1} w$. Write $P = \text{fix}_{k-1}(u, w)$ and $n = n_{k-1}(u, w)$. Set $a_0 = u^{-1}(w(k))$; since $u(k) \neq w(k)$ and $w(k)$ is a value of u occurring at a position $< k$ (this holds whenever the diagonal is nonzero), we have $a_0 < k$. The Monk summand indexed by t_{a_0k} then satisfies $u \xrightarrow{k-1} wt_{a_0k}$ with

$$\text{fix}_{k-1}(u, wt_{a_0k}) = P \cup \{w(k)\}, \quad n_{k-1}(u, wt_{a_0k}) = n - 1,$$

carries weight $q_{k-1}(u, w)$, and is the *only* Monk summand in $\mathcal{G}_M$. Dividing $\mathcal{G}_M$ by the common factor $q_{k-1}(u, w)$, its two contributions read

$$(y_{w(k)} - z_1) \mathbf{E}(y_P; z) - \mathbf{E}(y_{P \cup \{w(k)\}}; z),$$

which vanish since $\mathbf{E}(y_{P \cup \{w(k)\}}; z) = (y_{w(k)} - z_1) \mathbf{E}(y_P; z)$ by adjoining the variable $y_{w(k)}$. Hence $F_M = 0$.

Case (ii): $\mathcal{G}_M$ does not contain the diagonal. Then $q^M \neq q_{k-1}(u, w)$, and $\mathcal{G}_M$ consists of Monk summands of (6.3)–(6.4) together with the quantum term when $q^M = q_{k-1}q_{k-2}(u, w)$. By Lemma 6.12 these summands sum to 0, so $F_M = 0$.

As the two cases are exhaustive, every $F_M = 0$, so $F = 0$. $\square$

We now record the recurrence satisfied by the quantum factorial elementary polynomials, which drives the induction.

Lemma 6.14. *For all integers k we have*

$$\tilde{E}_{k,k}(x; z) = (x_k - z_1) \tilde{E}_{k-1,k-1}(x; z) + q_{k-1} \tilde{E}_{k-2,k-2}(x; z),$$

with the conventions $\tilde{E}_{0,0}^q = 1$ and $\tilde{E}_{k,k} = 0$ if $k < 0$. Moreover, the ordinary polynomials $\mathbf{E}(y_A; z) = \prod_{i \in A} (y_i - z_1)$ satisfy, for any set A and any $c \notin A$, the single-variable identity

$$\mathbf{E}(y_{A \cup \{c\}}; z) = (y_c - z_1) \mathbf{E}(y_A; z).$$

The first recurrence is the quantization of the classical relation $E_{k,k} = (x_k - z_1)E_{k-1,k-1}$; the extra term $q_{k-1} \tilde{E}_{k-2,k-2}$ is the only feature absent from the non-quantum setting. Discarding all quantum terms recovers the classical Pieri formula [8, Theorem 7.1], whose term-by-term cancellation is carried out in [8, Proof of Proposition 4.4] by means of the reflection-matching Lemmas 7.6–7.9 of that paper; we adapt that argument, the essential new points being the length-decreasing steps permitted by Lemma 6.3 and the attendant bookkeeping of the powers of q .

Proof of Theorem 3.2. For a permutation w , write $[\tilde{\mathfrak{S}}_w]F$ for the coefficient of $\tilde{\mathfrak{S}}_w(x; y)$ in an expansion F of a quantum double Schubert polynomial in the basis $\{\tilde{\mathfrak{S}}_w(x; y)\}$. With this notation the theorem is the assertion that, for all u and all $k \geq 1$,

$$[\tilde{\mathfrak{S}}_w](\tilde{\mathfrak{S}}_u(x; y) \tilde{E}_{k,k}(x; z)) = \begin{cases} q_k(u, w) \mathbf{E}(y_{\text{fix}_k(u, w)}; z) & \text{if } u \xrightarrow{k} w, \\ 0 & \text{otherwise.} \end{cases}$$

For brevity, when $u \xrightarrow{m} w$ we set

$$c_m(u, w) = q_m(u, w) \mathbf{E}(y_{\text{fix}_m(u, w)}; z),$$

and $c_m(u, w) = 0$ if $u \not\xrightarrow{m} w$. The claim is that $[\tilde{\mathfrak{S}}_w](\tilde{\mathfrak{S}}_u \tilde{E}_{k,k}) = c_k(u, w)$.

We argue by strong induction on k .

Base case $k = 1$. We have $\tilde{E}_{1,1}(x; z) = x_1 - z_1$, and Lemma 6.1 with $i = j = 1$ gives

$$(x_1 - z_1) \tilde{\mathfrak{S}}_u(x; y) = (y_{u(1)} - z_1) \tilde{\mathfrak{S}}_u(x; y) + \sum_{\substack{b > 1 \\ \ell(ut_{1b}) = \ell(u) + 1}} \tilde{\mathfrak{S}}_{ut_{1b}}(x; y) + \sum_{\substack{b > 1 \\ \ell(ut_{1b}) = \ell(u) - 2(b-1) + 1}} q_{1b} \tilde{\mathfrak{S}}_{ut_{1b}}(x; y).$$

The permutations w with $u \xrightarrow{1} w$ are exactly $w = u$ and $w = ut_{1b}$ for $b > 1$ with $u \triangleleft ut_{1b}$, since Definition 3.1 permits at most one reflection when $k = 1$. For $w = u$ we have $q_1 = 1$, $\text{fix}_1(u, u) = \{u(1)\}$, and $\mathbf{E}(y_{\{u(1)\}}; z) = y_{u(1)} - z_1$, matching the diagonal term. For $w = ut_{1b}$ we have $\text{fix}_1 = \emptyset$ and $\mathbf{E}(y_\emptyset; z) = 1$; if the step is length-increasing then $q_1(u, w) = 1$, matching the second sum, and if it is length-decreasing then $q_1(u, w) = q_{1b}$, matching the third sum. All other coefficients vanish on both sides. This proves the base case.

Inductive step. Fix $k > 1$ and assume the claim at all levels $m < k$. By Lemma 6.14,

$$(6.1) \quad \tilde{\mathfrak{S}}_u \tilde{E}_{k,k} = (x_k - z_1)(\tilde{\mathfrak{S}}_u \tilde{E}_{k-1,k-1}) + q_{k-1} \tilde{\mathfrak{S}}_u \tilde{E}_{k-2,k-2}.$$

By the inductive hypothesis, $\tilde{\mathfrak{S}}_u \tilde{E}_{k-1,k-1} = \sum_v c_{k-1}(u, v) \tilde{\mathfrak{S}}_v$. Applying Lemma 6.1 with $i = k$, $j = 1$ to each $\tilde{\mathfrak{S}}_v$ and extracting the coefficient of $\tilde{\mathfrak{S}}_w$ from (6.1), we obtain

$$(6.2) \quad [\tilde{\mathfrak{S}}_w](\tilde{\mathfrak{S}}_u \tilde{E}_{k,k}) = (y_{w(k)} - z_1) c_{k-1}(u, w)$$

$$(6.3) \quad + \sum_{\substack{b > k, \, wt_{kb} \triangleleft w \\ \ell(w) = \ell(wt_{kb}) + 1}} c_{k-1}(u, wt_{kb}) + \sum_{\substack{b > k, \, wt_{kb} \triangleleft w \\ \ell(w) = \ell(wt_{kb}) - 2(b-k) + 1}} q_{kb} c_{k-1}(u, wt_{kb})$$

$$(6.4) \quad - \sum_{\substack{a < k, \, wt_{ak} \triangleleft w \\ \ell(w) = \ell(wt_{ak}) + 1}} c_{k-1}(u, wt_{ak}) - \sum_{\substack{a < k, \, wt_{ak} \triangleleft w \\ \ell(w) = \ell(wt_{ak}) - 2(k-a) + 1}} q_{ak} c_{k-1}(u, wt_{ak})$$

$$(6.5) \quad + q_{k-1} c_{k-2}(u, w),$$

where we used that $\text{sign}(b - k) = 1$ for $b > k$ and $\text{sign}(a - k) = -1$ for $a < k$, and that a summand in (6.3)–(6.4) is present precisely when $u \xrightarrow{k-1} wt_{kb}$, respectively $u \xrightarrow{k-1} wt_{ak}$ (otherwise $c_{k-1} = 0$). We must show this equals $c_k(u, w)$. We distinguish three cases according to whether $u(k) = w(k)$ and, when $u(k) \neq w(k)$, whether $u \xrightarrow{k} w$; by Lemma 6.9 these cases are exhaustive and, in particular, the combination $u(k) \neq w(k)$ with both $u \xrightarrow{k} w$ and $u \xrightarrow{k-1} w$ never occurs.

Case 1: $u(k) = w(k)$. By Lemma 6.9(i) we have $u \xrightarrow{k} w$ if and only if $u \xrightarrow{k-1} w$; suppose first that both hold. Since $u(k) = w(k)$,

$$\text{fix}_k(u, w) = \text{fix}_{k-1}(u, w) \cup \{u(k)\}, \quad q_k(u, w) = q_{k-1}(u, w),$$

the second equality holding because no reflection with first index k occurs. The diagonal contribution (6.2) then gives

$$(y_{u(k)} - z_1) c_{k-1}(u, w) = q_{k-1}(u, w) (y_{u(k)} - z_1) \mathbf{E}(y_{\text{fix}_{k-1}(u, w)}; z) = q_k(u, w) \mathbf{E}(y_{\text{fix}_k(u, w)}; z) = c_k(u, w),$$

by adjoining the variable $y_{u(k)}$ as in Lemma 6.14. The remaining contributions are the reflection sums (6.3), (6.4) and the quantum term $q_{k-1} c_{k-2}(u, w)$ of (6.5), which sum to 0 by Lemma 6.10. Hence the total is $c_k(u, w)$, as required. If instead $u(k) = w(k)$ but $u \not\xrightarrow{k} w$ (equivalently $u \xrightarrow{k-1} w$), then $c_k(u, w) = 0$ and the diagonal term vanishes, while the reflection sums and quantum term again cancel by Lemma 6.10; the total is $0 = c_k(u, w)$.

Case 2: $u \xrightarrow{k} w$ but $u \not\xrightarrow{k-1} w$. By Lemma 6.8 there is a unique $b > k$ with $wt_{kb} \triangleleft w$ and $u \xrightarrow{k-1} wt_{kb}$; set $v = wt_{kb}$. This is exactly the unique nonvanishing summand of (6.3), and by Lemma 6.8 the sums (6.4) vanish, as does the diagonal (6.2) and the quantum term (6.5) (since $u \not\xrightarrow{k-1} w$ and $u \not\xrightarrow{k-2} w$). Moreover $\text{fix}_k(u, w) = \text{fix}_{k-1}(u, v)$, so

$$\mathbf{E}(y_{\text{fix}_k(u, w)}; z) = \mathbf{E}(y_{\text{fix}_{k-1}(u, v)}; z).$$

If $\ell(w) = \ell(v) + 1$ then the reflection t_{kb} is length-increasing, so by Definition 3.1 $q_k(u, w) = q_{k-1}(u, v)$ and the contribution is $c_{k-1}(u, v) = c_k(u, w)$. If instead $\ell(w) = \ell(v) - 2(b - k) + 1$ then t_{kb} is length-decreasing, Definition 3.1 gives $q_k(u, w) = q_{k-1}(u, v) q_{kb}$, and the contribution is $q_{kb} c_{k-1}(u, v) = c_k(u, w)$. In both cases the coefficient equals $c_k(u, w)$, as required. This is the step at which the quantum drop terms of Lemma 6.1 and the recursive definition of $q_k(u, w)$ interlock.

Case 3: $u \not\xrightarrow{k} w$ (with $u(k) \neq w(k)$). Here $c_k(u, w) = 0$, so we must show the coefficient (6.2)–(6.5) vanishes. This is exactly the assertion of Lemma 6.13. (The subcase $u(k) = w(k)$ with $u \not\xrightarrow{k} w$ was already treated at the end of Case 1.)

In all cases $[\tilde{\mathfrak{S}}_w](\tilde{\mathfrak{S}}_u \tilde{E}_{k,k}) = c_k(u, w)$, completing the induction and the proof. $\square$

7. ADDITIONAL FORMULAS

Definition 7.1. Consider a permutation $d_2[k]$ satisfying

$$\mathfrak{c}(d_2[k]) = \overbrace{(2, 2, \dots, 2)}^{k \text{ times}}.$$

The double Schubert polynomial $\mathfrak{S}_{d_2[k]}(x; y)$ is a product of two factorial elementary symmetric polynomials, specifically:

$$\mathfrak{S}_{d_2[k]}(x; y) = E_k(x; y_1) E_k(x; y_2).$$

Therefore, the Pieri formula in [8] easily transfers to being able to multiply by $\mathfrak{S}_{d_2[k]}(x; y)$.

For quantum double Schubert polynomials, $\tilde{\mathfrak{S}}_{d_2[k]}(x; y)$ is *not* a product of factorial quantum elementary polynomials. For example:

$$\tilde{\mathfrak{S}}_{d_2[1]}(x; y) = \tilde{E}_{1,2}(x; y) \tilde{E}_{1,1}(x; y) - E_{2,2}^q(x; y)$$

$$\tilde{\mathfrak{S}}_{d_2[2]}(x; y) = E_{2,3}^q(x; y) E_{2,2}^q(x; y) - E_{3,3}^q(x; y) E_{1,2}^q(x; y)$$

For abbreviation purposes, we denote this polynomial by $D_k^q(x; y)$.

We define a relation $\overset{k}{\rightrightarrows}$ on permutations as follows. We say that $u \overset{k}{\rightrightarrows} w$ if and only if there exists a $v \in S_\infty$ such that $u \overset{k}{\rightsquigarrow} v$ and $v \overset{k}{\rightsquigarrow} w$ satisfying the additional condition that if $u^{-1}v$ is a product of disjoint cycles $c_1, \dots, c_m$ and $v^{-1}w$ is a product of disjoint cycles $d_1, \dots, d_n$, then for each d_i there exists a c_j such that $d_i \cap c_j = \{\max(d_i)\} = \{\max(c_j)\}$. Call the set of all such v $\text{pivot}_k(u, w)$.

Then

Theorem 7.2. *We have that the product $\tilde{\mathfrak{S}}_u(x; y) D_k^q(x; z)$ is equal to*

$$\sum_{\substack{k \\ u \overset{k}{\rightrightarrows} w}} \sum_{v \in \text{pivot}_k(u, w)} q_k(u, v) q_k(v, w) E_{k-n_k(u, v)}(y_{\text{fix}_k(u, v)}; z_1) E_{k-n_k(v, w)}(y_{\text{fix}_k(v, w)}; z_2) \tilde{\mathfrak{S}}_w(x; y).$$

8. PETERSON-WOODWARD COMPARISON FORMULA

Proposition 8.1 ([4, Proposition 2.10]). *Let W_P be a parabolic subgroup of W . Then for any $\lambda_P \in Q^\vee / Q_P^\vee$, we have that*

$$c_{u,v}^{w, \lambda_P}(y; y) = c_{u,v}^{w \omega_P \omega_{P'}, \lambda_B}(y; y),$$

where ω_P is the longest element in W_P , $\omega_{P'}$ is the longest element in $W_{P'}$, where P' is the parabolic subgroup generated by $\{s_\alpha \mid \alpha \in \Delta_P, \langle \alpha, \lambda_B \rangle = 0\}$, and λ_B is the Peterson-Woodward lift of λ_P .

Corollary 8.2. *The Peterson-Woodward comparison formula holds for $c_{u,v}^w(y; z)$.*

Proof. We have that

$$c_{u,v}^{w, \lambda_P}(y; z) = \sum_{v_1 \cdot v_2 = v} c_{u, v_2}^{w, \lambda_P}(y; y) \mathfrak{S}_{v_1}(y; z)$$

By definition of a parabolic subgroup, $v_2 \in W_P$ as well. Thus, this is equal to

$$\sum_{v_1 \cdot v_2 = v} c_{u, v_2}^{w \omega_P \omega_{P'}, \lambda_B}(y; y) \mathfrak{S}_{v_1}(y; z) = c_{u,v}^{w \omega_P \omega_{P'}, \lambda_B}(y; z).$$

$\square$

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